

Review article

RF/WiFi-based UAV surveillance systems: A systematic literature review

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ABSTRACT

The deployment of Unmanned Aerial Vehicles (UAVs) has gained increasing interest in various applications during recent years due to its multiple features. At the same time, the misuse of UAVs has also posed serious concerns related to privacy, safety, and security of individuals and organizations. This has caused the utmost demand for drone surveillance systems to monitor suspicious activities of any UAVs. In this context, this paper aims at presenting a systematic literature review of the most interesting works concerning the design and development of drone surveillance systems for target detection and tracking based upon RF/WiFi techniques. More specifically, the research tackles the addressed design issues and challenges, the employed methods and techniques. Furthermore, the methodology of the literature review is carefully described, and the performance assessment of relevant studies is explored and analyzed, based on specific criteria. Provided the analysis of state-of-the-art, future research trends are also highlighted in detail, in order to drive the scientific community to research and design novel and efficient drone surveillance systems based on RF/WiFi techniques.

1. Introduction

Low cost, small size, ease of availability, and especially live video feed have provided great popularity to drones in the latest years since they are being used in a variety of applications [1], such as in [2,3]. Also, UAVs play a crucial role in the connected smart cities of the future [4]. Unfortunately, the continuous increment in popularity of drone applications has posed safety, security, and privacy concerns, mainly due to the so-called intelligent attacks and threats [5–7]. Different incidents are observed in the recent past due to the misuse of drones. For example, a drone was dropped on a subway track in Singapore in 2015, near the international airport of Los Angeles a Lufthansa air-jet came within 200ft of colliding with a drone and recently, in 2019, two drones attacked the oil fields of Saudi Arabia, interrupting the main pipeline across-country [8,9]. Other incidents observed between 2014 and 2016 include unauthorized photography, building and structural collision, and many other events mentioned in [10]. Such accidents reflect the paramount importance of the availability of drone surveillance systems in nowadays society.

According to Federal Aviation Administration (FAA), over a hundred calls regarding the presence of drones in the visible proximity of air-crafts in the vicinity of airports were reported by the commercial pilots [11]. Furthermore, the number of Unmanned Aircraft System (UAS) sales in the commercial sector was estimated to 400,000 in 2019, an increase in the growth rate of 44% over its rate in 2018 and the exact number of these vehicles was reported as 385,000 at the end of 2019, according to the FAA report [12]. This continuous increase in drones is alarming, since it increases the chances of collisions with commercial air-crafts at airports.

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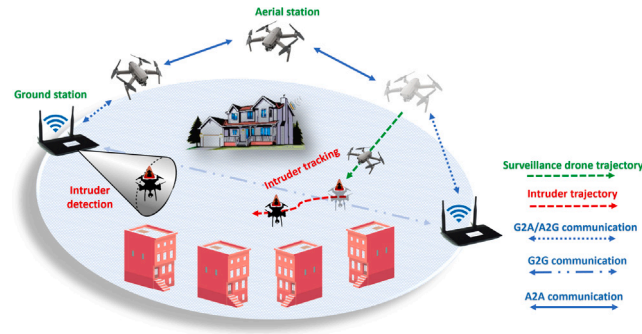


Fig. 1. A conceptual diagram of a typical drone surveillance system.

Table 1

Manufacturing companies for drone surveillance system.

Company Name	Detection	Localization	Tracking	Techniques	Detection Range
Dedrone	✓	✓	✓	RF, Video, Audio	≤ 1 Mile
Drone Detector	✓	×	×	RF, Audio, Video, Thermal, Radar	≤ 1000 m
Dronesshield	✓	✓	✓	RF, Radar, Audio	≤ 40 m
Orella Drone Detector	✓	✓	✓	Audio	≤ 100 m
DD Counter Measures	✓	×	×	RF	≤ 15 m

Moreover, not only in the USA, but globally, air traffic density and complexity are increasing. Such proliferation of UAVs enables the mixed airspace for both the UAVs and national airspace [13]. This requires the development and integration of efficient and reliable communication, navigation and surveillance system to ensure safe and reliable operations among different entities like ground stations and aircraft. Although different regions make regulations in the flying/non-flying of UAVs in specific areas and zones, people must comply with such rules, and a violation of formal regulations is intended as crime or felony. Fig. 1 presents a conceptual view of a typical drone surveillance system, which uses aerial and ground surveillance stations to detect and track potential intruders in a specific area. Specifically, a ground surveillance station has detected one intruder, while an aerial surveillance station is tracking the other. An intruder, a malicious drone, is always non-cooperative and attempts to conduct malicious activities. Between ground and aerial stations, communication links such as Air-to-Ground (A2G), Ground-to-Air (G2A), Ground-to-Ground (G2G), and Air-to-Air (A2A) can be used to optimize the surveillance system's effectiveness for detection and tracking tasks.

Above highlighted incidents and scenarios raised the utmost necessity to design and develop accurate and efficient anti-drone (or counter-drone) systems. An important characteristic of drone surveillance systems is the implementation of diversified features such as detecting authorized/unauthorized drones, classifying different drones, localizing and tracking the intruders in different fly/no-fly zones. In any case, basic safety precautions and measures shall be timely adopted in order to avoid any potential threat or danger. Therefore, multiple researchers from academia and industries have tried to develop proper drone surveillance systems. Nevertheless, there are still a number of issues to be investigated, and researchers are still proposing novel solutions, trending to provide also small scale implementations to the benefit of individuals.

Due to many threats and incidents observed in the recent past, several companies have also been engaged in manufacturing counter-drone systems, producing commercial drone surveillance systems mainly addressing the military framework, even though some of them cover the civil domain as well. Most of the companies use multiple receivers by integrating different drone detecting techniques to make the system more reliable [14], thus making them unfeasible solutions for the deployment in the domestic domain [15]. A comparison of drone surveillance systems manufactured by different companies including the provided features and capabilities is summarized in Table 1. Information about the characteristics of the shortlisted systems are obtained from [16], as well as from the respective companies websites [17–20].

1.1. Drone surveillance system

Public safety, national security but also citizens' privacy are critical concerns in the framework of UAV deployment, since remote piloted vehicles might represent a threat when violating flying regulations. Typically, four main challenges have been considered broadly by many researchers contributing to this field: drone detection, localization, tracking, and controlling or deactivation of the drones. Devising a surveillance system able to provide all these features and to guarantee significant accuracy is a big challenge. In this framework, the present Systematic Literature Review (SLR) is carried out for drone detection and tracking systems topic only. The main motivation is that the primary and fundamental step in drone surveillance is the detection of the malicious drone. Furthermore, to interdict the identified malicious drone, it is not sufficient to locate it, yet mandatory to track it. Hence, tracking UAVs is also a critical issue in surveillance systems, being it also a hard task to perform, since the behavior of drones might be highly unpredictable. It is important to track the intruder until countermeasures are applied, or the intruder is neutralized.

1.2. RF/WiFi techniques

In RF and WiFi-based techniques, features are extracted at physical and data link layers, respectively. RF fingerprints depend upon the unique characteristics of the RF signal captured either from different drones or their remote controllers. Several experiments have been performed in the literature showing that different drones have unique features due to multiple reasons, such as their shape, circuitry and modulation techniques. Hence, these unique signatures have been used as a reference for drone detection and classification [21]. Similarly, WiFi fingerprinting techniques employ specific RF sniffers, which eavesdrop the drone communication signals to and from the controller or base station, including both control and video transmissions [22]. Packets included in captured WiFi transmission are less affected by the obstacles and other flying objects. The information regarding drone type and their operation modes, can also be extracted from WiFi traffic where the communication is not encrypted. Such cyber information can also be used in forensic investigations. The WiFi-based approach has been proposed in recent years, since the availability of WiFi-operated drones in the market has dramatically increased. Nonetheless, for what concerns the literature included in the shortlisted articles, works presenting drone surveillance systems based on RF techniques are almost double the works proposing WiFi-based approaches.

1.3. Relevant work

Different survey reports and short reviews have been carried out in the context of drone surveillance systems. Literature works focusing on aerial stations are provided in [23,24], while [7,13] discuss both ground and aerial-based stations. A market survey report is presented in [25], which investigates the challenges regarding the detection of Low, Slow, and Small (LSS) UAVs and reports the main classification and neutralization approaches implemented so far. Concerning the localization of malicious drones, [26] presented a comprehensive survey for the different localization techniques and possible approaches available in the literature that can be used to locate any non-cooperative drones. Furthermore, authors in [24] discuss signal strength based localization and reviewed the different modalities of drone detection techniques, including video, audio, Radar, and RF, by presenting their advantages and disadvantages. A short review of related works, including crucial technologies regarding detection and tracking of drones relying only on monitoring drones, is presented in [23], where Flying Ad-hoc Network (FANET) architectures for the deployment of monitoring drones for detection, localization and tracking are discussed along with their limitations and insights on future directions. Similarly, [27] also discussed anti-drone systems from a detection, localization, and tracking perspective, covering both ground and aerial stations using RF and WiFi approaches.

In addition, techniques for the detection and interdiction of small UAVs are reviewed and investigated in [13]. Authors mainly focus their attention on techniques based on Radar. However, other techniques, including acoustic, vision, and RF, were also outlined. In this connection, a similar analysis is presented in [7], where authors summarize possible ways for detecting and tracking the drones along with feasible approaches to interdict malicious targets. In addition, cyber and physical threats caused by drones are highlighted and advantages/disadvantages of the surveyed detecting techniques are summarized. Moreover, [28,29] provides an overview of the state of the art aimed at only detecting UAVs. Implementation of ML techniques for the detection and classification of malicious drones is also surveyed in [30]. This survey mainly focuses on the detection and classification aspect of drones by considering different technologies, including Radar, image, sound, and RF. Moreover, counter UAV detection systems based upon radar, acoustic, thermal and optical cameras, and systems relying on the fusion of multiple types of sensors using Deep Learning (DL) approaches are reviewed in [31].

1.4. Contributions

As reported above, so far the state of the art has emphasized Radar, video, and acoustic approaches rather than RF and WiFi techniques. Along with ground-based surveillance, aerial surveillance of intruders has been studied. Additionally, different reviews have been provided regarding the implementation of drone detection systems based on ML and DL algorithms, with an emphasis on developed algorithms. Given these considerations, our survey distinguishes from other studies in the literature by providing the following novel contributions:

- The main focus of the present survey consists in investigating RF and WiFi-based drone surveillance systems for detection and tracking of UAVs.
- This study provides a systematic way of understanding the addressed issues, procedures, methods, techniques, algorithms and experimental setups being carried out in the literature, along with a performance assessment analysis.
- In-depth design issues, open challenges, and future trends in the development of drone surveillance systems are highlighted and discussed in detail.

This survey work is organized as follows: Section 2 presents the SLR process and the demographic data obtained after following the SLR process. The surveyed papers are presented following specific research questions, and further described in the subsequent sections from both the detection and tracking point of view. Design issues and challenges are discussed in Section 3, while methods and procedures adopted by the different works are described in Section 4. In addition, performance analysis and effectiveness of these systems have been discussed in Sections 5 and 6, respectively. Future research directions are presented in Section 7. Finally, in the last section the SLR is summed up by drawing the conclusions. Fig. 2 depicts the global picture of the structure of this survey work.

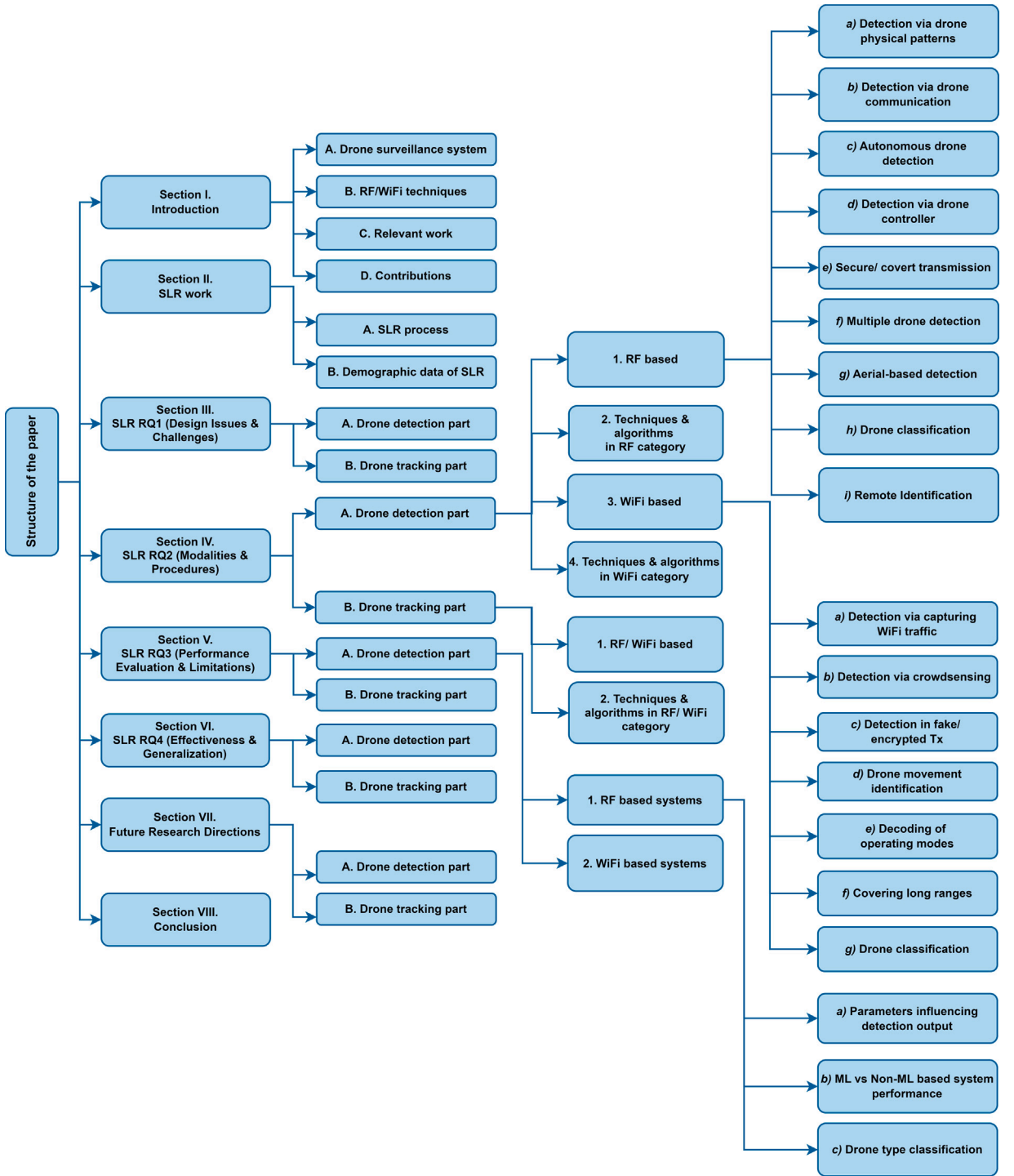


Fig. 2. Global picture of the paper.

Table 2
Summary of information sources and paper selection criteria.

Parameter	Description
Electronic Databases	IEEE Xplore, Springer Link, Elsevier, Google Scholar
Search items	Journals, Conference Proceedings, Theses/Dissertations
Publication Period	Jan 2010 - Dec 2022
Language	English

2. SLR work

Recently, the investigation on surveillance systems in the general and anti-drone systems in particular, has been a topic of great interest due to an increase in the number of security issues caused by UAVs, and this has triggered the need to come up with a systematic review of the literature in this specific field.

2.1. SLR process

A definition of the correct methodology to conduct the SLR work has been provided in [32]. In addition, a specific process to carry out a systematic review work is also given in [33]. In detail, the main steps to be considered are:

- Plan the review
- Conduct the review
- Document the review

Our review work has been carried out by following the guidelines and procedures described in the aforementioned references, in order to produce a systematic survey of the literature. This section has the primary objective of devising the research questions, and hence to describe in detail the adopted review protocol. We also discuss possible conflicts and biases and how to overcome such issues, the searching approach, and the criteria employed for article inclusion/exclusion.

2.1.1. Formulation of research questions

Provided the importance and need of proper drone surveillance systems, this systematic review begins with the identification of studies related to the detection and tracking issues inherent of a drone monitoring platform. These could be considered the fundamental features of any system aimed at interdicting malicious targets. In order to implement such sub-systems, different techniques have been considered so far, such as audio, video, RF, WiFi, Radar. As a matter of fact, the peculiarity of the present SLR consists in focusing specifically on RF/WiFi based approaches. In order to carry out the review, the following research questions (RQs) have been devised, related to the defined objectives.

RQ1: Which design issues have been addressed?

RQ2: Which modalities and procedures have been employed for both drone detection and tracking?

RQ3: What is the performance of such sub-systems (accuracy, constraints and limitations).

RQ4: Are the considered methods effective and versatile?

Each RQ is designed to extract and categorize data pertaining to drone detection and tracking systems. In particular, the aim is to extract information about the addressed challenges, the employed techniques and algorithms, the adopted methodologies, the observed limitations and constraints, and the obtained performance. The final component of these RQs examines the versatility of the existing systems in relation to the ability of adapting to different scenarios, such as the working environment, the type of drones, and the implemented methods.

2.1.2. Review protocol

Developing the review protocol eliminates researcher bias and promotes review quality and reproducibility. Following the procedure proposed in [33], the review protocol is to be defined explicitly after devising the research questions, so that a bias- and conflict-free systematic review may be carried out. To this aim, we begin by listing the sources of information to be explored, followed by the search strategy, the definition of inclusion and exclusion criteria for literature works, and finally describing the data extraction process.

Information source. Electronic databases are used to collect the research studies to be reviewed. A full electronic search strategy for at least one database needs to be carried out following the guidelines in [32]. In this connection, *IEEE Xplore* has been used as the main search database, but also other sources have been considered, including *Elsevier*, *Springer*, and *Google Scholar*. The reason for including a scientific search engine such as *Google Scholar* is to cover any article related to the survey works not included in the aforementioned publisher databases, namely other publishers' journals and conference proceedings, Ph.D. and Master thesis. A summary of the considered information sources and paper selection criteria is provided in [Table 2](#).

Eligibility criteria. The literature work contributing to the drone surveillance framework has been carried out mainly in the last decade, hence we have considered as reference the years between 2010 and 2022. In this regard, research studies that cover either one or all of the formulated research questions and available in the specific sources mentioned in Table 2 have been considered eligible to be included in the present review, by selecting only English-written works. In addition, the articles shall also comply with the adopted inclusion/exclusion criteria described in the following.

Search criteria. Search criteria are defined following the guidelines in [32,33], applying them individually to the drone detection and tracking sub-topics. In order to find the best possible matches among literature works, specific keyword searches and their synonyms are used. Primary studies have been selecting with a two-stage process by searching specific keywords connected with proper Boolean expressions (*i.e.*, AND/OR). In the first part, all the studies related to drone detection are explored and shortlisted, while drone tracking-related studies are identified in the second part.

As far as drone detection is concerned, the specific search keywords used in the databases are: Drone detection *OR* UAV detection, Drone monitoring *OR* UAV monitoring, Drone surveillance *OR* UAV surveillance, and drone identification *OR* UAV identification. Similarly, for what concerns drone tracking, sample keywords and their combinations are: UAV *AND* tracking, aerial target *AND* tracking, drone *AND* tracking and UAV following *OR* drone following. All keywords were queried in *IEEE Xplore*, since it is the main scientific source database. However, specific keywords and expressions were also queried in the rest of the considered databases.

The query with all relevant keywords provided a quite large number of articles, yet those related to the research questions were relatively less. To cope with this situation, a stop criterion was applied after going through the first hundred of articles collected with the primary keyword search. This stop criterion was explicitly applied to the databases, except for *IEEE Xplore*. Furthermore, search filters were applied, determined by the eligibility criteria, *e.g.*, publication year, language and type of articles.

Inclusion/exclusion criteria. Part of the collected articles was not related to the considered research questions, which requires to set inclusion and exclusion criteria in order to select the most relevant papers and studies for the secondary research. Eligibility criteria translate into search filters on the databases. Moreover, inclusion/exclusion criteria are used to extract only the most relevant works from the pool of studies obtained from the database search. Therefore, the following exclusion criteria have been employed in this SLR:

- i Secondary work: Other existing survey and review articles are considered secondary work, since they do not contribute directly to designing and developing anti-drone systems. Though they are not considered to be included as primary research, they have been included for comparison purposes with the present survey.
- ii Not related to UAVs: Searched articles and papers are not included if they address targets other than UAVs, such as ground objects, even if they address detection and tracking purposes.
- iii Commercial applications: Studies related to the utilization of UAVs for commercial purposes are not considered to be included in this survey.
- iv Cooperative UAV and RF sources: In the anti-drone system, target UAVs and drones are always non-cooperative. Therefore any study related to detection and tracking in which UAVs are cooperative, are excluded.
- v Techniques other than RF/WiFi: Drone surveillance systems implemented with techniques other than RF/WiFi were all excluded from the review process.
- vi Aerial source and ground objects: Work carried out for monitoring and detecting ground objects with the help of aerial stations are also excluded, since counter-drone systems are usually devised to work in the reverse way (*i.e.*, identify flying targets from the ground).
- vii Other applications of UAVs: Utilization and deployment of UAVs is applied to multiple fields (*e.g.*, delivery systems, search and rescue operations); works related to such applications is out of the scope of this research work.
- viii Countermeasures for Intruders: Concept of RF signals jamming, GPS spoofing, and related techniques and methods implemented to tackle any intruder, either ground or aerial and fixed or moving, are also not considered.
- ix Cooperation and coordination of UAV Swarms: UAVs are deployed also in swarms, where multiple devices cooperate for a common goal; in such case the focus is given on their communication and coordination with other devices and the ground stations. This survey keeps such works out of the context of the research theme.

The journal's rank, the relevancy of the article contents, technical novelty with the subject matter of the articles, publishing year, and finally, the perspective of the authors of this review work on the subject domain were also criteria followed for the inclusion of articles. Finally, the most relevant articles identified by the process described above have been included in the quantitative and qualitative analysis of the specific contents.

Quality criteria. Selection of the quality of research articles has a significant impact on the quality of the research. According to [33], quality criteria are not intended to include/exclude works from the primary research. On the other hand, they are employed to evaluate the quality of the selected articles. Conventionally, a list of key questions is formulated to set these criteria, and we carried out this step by following the guidelines in [33], as summarized in Table 3. The quality-related questions for the primary study are used to assess the standard of the research article; namely, a positive answer reflect a high quality standard of the considered work.

Article selection process. In this review, a two-step process is followed in the selection of a specific article.

- i. Screening based upon title and abstract
- ii. Selection of the most relevant articles based upon full-text analysis

Table 3
Quality criteria questions.

Qs No.	Quality Questions
01	Is the research study based on solid principles?
02	Does the article provide sound details of the addressed research context, along with motivations and explanations?
03	Does the research provide a clear description of the contribution and experimental results to support the findings?
04	Is there a thorough discussion about potential issues and limitations of the proposed work?

Initially, sample keywords are applied to the electronic search databases, at the same time ensuring the eligibility criteria. A relatively large number of articles and research studies are listed, including all publishers in that database. As a first step, among the shortlist of articles and research studies, the screening process based upon the title of the article and abstract is carried out to extract the relevant studies, while the rest of the articles are excluded from the review. In the second step, the articles collected from the first step are reviewed to assess the relevancy concerning the objectives of the secondary research. Inclusion/exclusion criteria are ensured during both of these steps.

Biases and conflicts. In order to eliminate biases, conflicts and disagreements, a specific methodology was applied for article selection and review. In particular, different tasks, paper eligibility and inclusion/exclusion, quality assessment, collection and extraction of data were initially performed by two authors. Hence, the resulting study was discussed in detail and verified by a third author in order to solve conflicts of viewpoints and overcome any biases. Finally, complete review and evaluation was carried out by the rest of the authors, thus leading to the proposed survey.

2.1.3. Data collection process and analysis of results

In the following, the data collection process is defined for the drone detection and tracking sub-systems, respectively. Two different data-extraction tables have been devised to extract the relevant information from the selected articles for each sub-system. The attributes considered in the first table are: (i) Reference (ii) Design issues, (iii) Method (iv) Techniques (v) ML/DL algorithms. The attributes chosen for second table are: (i) Reference (ii) Surveillance station (iii) Experiments/Simulations, (iv) Environment, (v) Drones/Controllers, (vi) Constraints (vii) Range, (viii) Outcome. Concerning tracking systems only, we considered one extra attribute, which pertains the nature of (ix) source signal, and the replacement of attribute (v) with the number of receivers. These attributes have been used to collect and efficiently summarize the information related to both drone detection and tracking sub-systems. The reviews of the shortlisted articles are presented separately for each sub-system (detection/tracking) to carry out a more specific and thorough analysis.

2.2. Demographic data of SLR

In this part, we discuss the implementation of the review protocol for information and data collection. As already mentioned in a set of keywords and phrases was used to search in the different databases, in order to comply with the considered search criteria described in the same section. While searching in databases, eligibility criteria, mentioned in , were followed strictly. The total number of articles retained from the *IEEE Xplore* were 988 and 896 for the drone detection and tracking systems, respectively. The overall quantity of retained articles is then 1884. As far as other databases are concerned, the number of articles obtained with the set of keywords and phrases was 127 and 196 for detection and tracking sections respectively, making the overall quantity of selected articles to 323. Hence, we followed the two-step study selection process mentioned in , where the articles are screened based on title and abstract first, causing the inclusion of 89 articles and exclusion of 2207 studies. This high number of excluded articles is due to the exclusion criteria mentioned in . The same exclusion criteria were also applied to the second step of the study selection process, where the articles were screened again based on the full-text reading. More specifically, the contents of the screened articles from the first step were studied and reviewed, and they were excluded only when any of the exclusion criteria were satisfied. In conclusion, the articles shortlisted for the analysis and critical review are 25 for what concerns drone detection and 3 about drone tracking.

The overall scheme of the employed review protocol is depicted in Fig. 3, according to the flow diagram proposed in [32]. Once the primary articles for the survey have been selected, each work was reviewed by two authors, and the third author also reviewed it in case of any conflict and disagreement between the authors. Hence, biases and conflicts were rectified, thanks to the procedure mentioned in Section “Review protocol (Biases and conflicts)”. Furthermore, questions related to quality criteria (see Table 3 in Section “Review protocol (Quality criteria)”) were discussed and answered for each considered article, thus ensuring not only the relevancy, but also the quality of the selected articles.

Fig. 5(a) shows the distribution of shortlisted studies for RF/WiFi-based drone surveillance systems published by different publishers. The large majority of studies (68%) were collected from the *IEEE Xplore* database, while only 4% of the articles are published in the MDPI library, which is the least percentage of articles among all considered publishers. Studies published by ACM are 14% of the total amount of selected articles, which is the same quantity belonging to Elsevier and others (namely, Internet society and University). In addition, the distribution of shortlisted articles, based on the type of publication, is presented in Fig. 5(b).

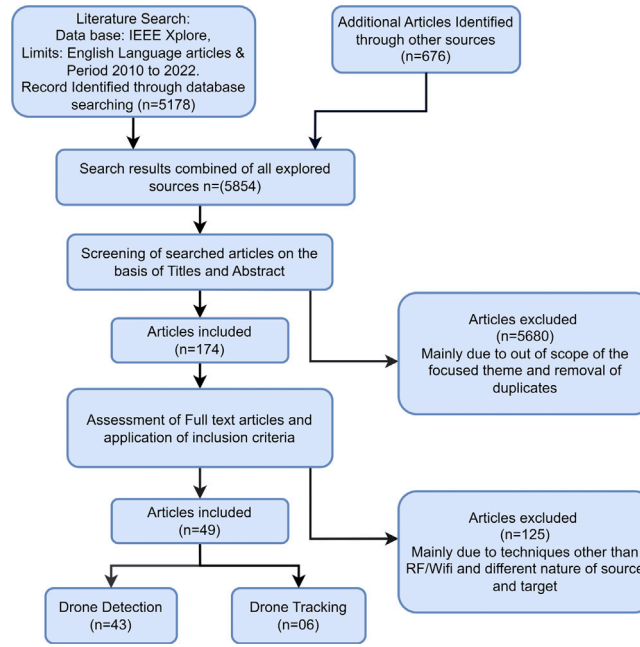


Fig. 3. Implementation of Review protocol.

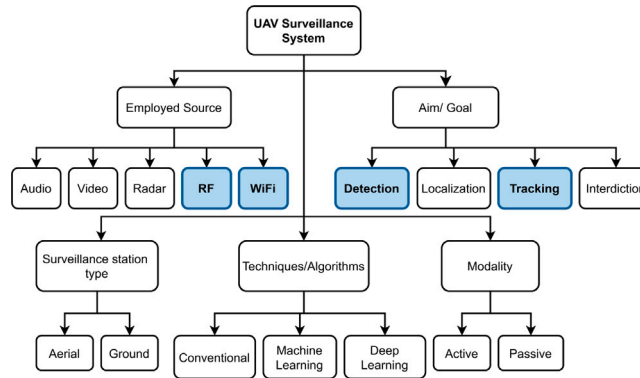


Fig. 4. Characteristics wise classification of drone surveillance system.

According to this distribution, slightly less than half of the selected articles, *i.e.*, 46%, are journal articles, since it was the main focus during the searching phase. Magazine, thesis, and symposium publications reach almost one-quarter of the total amount of articles, while one-third of all articles belong to conference proceedings, both IEEE and ACM. As a further analysis, a summary of the shortlisted articles is presented in Table 4 for both drone detection and tracking systems. A taxonomy of the drone surveillance systems proposed in the considered articles is provided in Fig. 4. The sub-blocks with solid bold borders (RF, WiFi, Detection, Tracking) under the umbrella of *Employed Source* and *Aim/Goal* blocks highlight the main focus of this study.

3. SLR RQ1 (Design issues and challenges)

This section addresses RQ1, which is related to the design issues, problems, and challenges under the umbrella of RF/WiFi techniques for drone detection and tracking systems. In order to answer this question, information about detection extracted from the reviewed studies are presented separately for both RF and WiFi techniques, as shown in Table 5 and Table 6, respectively. For

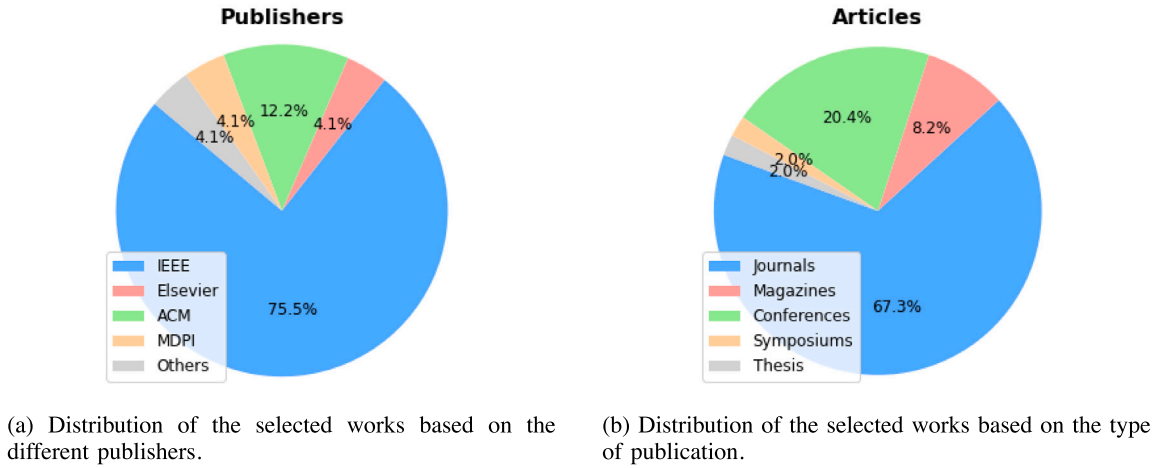


Fig. 5. Distribution of the shortlisted studies: based on the different publishers (left) and based on the type of publication (right).

what concerns drone tracking systems, the relevant information are summarized in Table 7. Based upon these insights, problems and design issues tackled by the research community are thoroughly discussed.

3.1. Drone detection part

In the following, we identify and highlight the main challenges related to the RF/WiFi-based detection techniques. Usually, drones employed in privacy attacks and threats are low cost, remotely controlled, and typically operating in the WiFi frequency band. The present study revealed the following major design issues and challenges, summarized in Tables 5 and 6:

- **RF receiver:** Since the strength of the RF receiver should be sufficient to capture the RF signals, the fundamental parameters that need to be investigated are related to receiver power, antenna gain, and similar quantities.
- **Range:** Designed system should be sensitive enough to capture the drone's communication, since signal power is relatively low and quickly lose strength over distance.
- **Multi-path Fading:** Many obstacles, especially in urban environments and densely populated areas, can affect the RF signal causing multipath fading. This effect usually leads to signal detriment, thus increasing the difficulty for the receiver to correctly and promptly detect the intruder's signal.
- **High Altitude:** As mentioned above, since the drone's signal strength is not very high, UAVs flying at altitude might make the drone identification task problematic, depending on the capturing ability of the designed system.
- **LoS & NLoS detection:** Designed system should include the capability to capture both Line of Sight (LoS) and Non Line of Sight (NLoS) drone signals, in order to be deployable in either rural or urban areas. This is a hard challenge, since these types of signals have different statistical properties.
- **Interference & Noise:** Normally, transmitted RF signals are subject to interference and noise; if the signal is too altered or degraded, identification of drone signatures in this condition can be quite problematic.
- **Detection time:** Detection systems should be designed in order to detect drones in almost real-time, since the device speed could cause the target to quickly disappear from the detection range.
- **Received signal decision:** After capturing a target's signal, the system must be able to promptly and efficiently determine whether it represents a threat or not, thus minimizing potential false alarms.
- **Selection of Hardware:** The choice of the hardware components to be employed for system development (*i.e.*, antenna, sensors, controlling circuitry, *etc.*) should be optimized in order to reach a trade off between efficiency and low production and maintenance costs. Although this consideration holds for any deployment domain, it becomes especially important when considering domestic applications.
- **Drone RF signal frequency:** Since differences in construction and hardware of UAVs translated into different transmission frequencies for the RF signal, it is a fundamental feature of any drone detection system to be able to correctly scan the different spectral bandwidths.
- **Authorized/Unauthorized drone:** Some areas impose the interdiction of any flying objects, while other zones establish a distinction between authorized/unauthorized devices. A proper drone detection system, given an identified target, should be able to decide if a drone is allowed to enter the area or shall be reported as an intruder.
- **Decoding of drone signal:** Depending on the detection technique and the drone communication protocol, it might be necessary to decode the captured signals in order to extract the information required to identify and neutralize potential threats.
- **Detection in covert/encrypted transmission:** The captured communication might be secured or encrypted. This makes the recognition process particularly challenging, forcing to adopt specific detection techniques able to identify the potential threats from the statistical features of the signal, without the need of plain-text transmissions.

Table 4
Summary of all shortlisted studies.

Ref.	Employed Signal		Aim/Goal		Techniques/Algorithms			Modality	
	RF	WiFi	Detection	Tracking	Conventional	ML	DL	Active	Passive
[15]		✓	✓		✓				✓
[21]	✓		✓		✓				✓
[22]		✓	✓			✓			✓
[24]	✓		✓				✓		✓
[34]	✓			✓	✓				✓
[35]	✓			✓	✓				✓
[36]	✓			✓	✓				✓
[37]	✓			✓	✓				✓
[38]	✓			✓	✓				✓
[39]	✓			✓	✓				✓
[40]	✓		✓			✓			✓
[41]	✓		✓		✓				✓
[42]	✓		✓		✓				✓
[43]	✓		✓		✓			✓	✓
[44]	✓		✓		✓				✓
[45]	✓		✓		✓				✓
[46]	✓		✓				✓		✓
[47]	✓		✓				✓		✓
[48]	✓		✓		✓				✓
[49]	✓		✓			✓			✓
[50]	✓		✓		✓				✓
[51]	✓		✓				✓		✓
[52]	✓		✓				✓		✓
[53]	✓		✓		✓				✓
[54]	✓		✓			✓			✓
[55]	✓		✓				✓		✓
[56]	✓		✓				✓		✓
[57]	✓		✓				✓		✓
[58]	✓		✓		✓				✓
[59]	✓		✓		✓				✓
[60]	✓		✓		✓				✓
[61]	✓		✓		✓				✓
[62]	✓		✓			✓			✓
[63]	✓		✓				✓		✓
[64]	✓		✓			✓			✓
[65]	✓		✓				✓		✓
[66]	✓		✓				✓		✓
[67]	✓		✓				✓		✓
[68]	✓		✓		✓				✓
[69]	✓		✓			✓			✓
[70]	✓		✓				✓		✓
[71]		✓	✓			✓			✓
[72]		✓	✓			✓			✓
[73]		✓	✓			✓			✓
[74]		✓	✓			✓			✓
[75]		✓	✓			✓			✓
[76]		✓	✓			✓			✓
[77]		✓	✓		✓				✓
[78]		✓	✓		✓				✓
	79.6%	20.4%	87.7%	12.3%	47%	26.55%	26.55%	3.57%	100%
	100%		100%		100%			100%	

3.2. Drone tracking part

In the following, we identify and discuss the main design issues and challenges addressed so far while designing drone tracking systems. In this connection, sources are reported in Table 7. In the considered studies, tracking of the intruders has been carried out by considering multiple information about drone localization. When dealing with drone surveillance systems, the target is considered as non-cooperative and this opens the door to investigate techniques to find the direction and physical position of the intruders. Tracking the moving targets is critical: designed algorithms should not be too complex or too slow, as this could lead to large estimation errors due to the aerial moving target not hovering and possibly escaping the monitoring region. Without a fast scanning phase, location and tracking of targets becomes impossible. In this framework, both ground and aerial-based tracking studies are available in the literature. The system design is extremely variable for both ground and aerial-based surveillance stations. It can be observed from the Table 7 that design issues discussed in [34,35] tackle general challenges in tracking the UAVs, based upon localization techniques. On the other hand, [36] consider a more practical approach, tackling the design issue related to the

problem of tracking intermittent UAVs. Considering drones with intermittent activity provides a novelty in drone tracking scenarios, contributing in the diversity of drone surveillance systems.

4. SLR RQ2 (Modalities and procedures)

In this section, RQ2 is addressed by providing the details of modalities and procedures for designing and developing the detection and tracking systems in the framework of drone surveillance systems.

4.1. Drone detection part

In literature, the term identification is used interchangeably for drone detection and classification tasks. For instance, drone detection is the identification of the received RF signal as a drone signal [41], and drone classification is the identification of the drone type and its operating modes [47]. Furthermore, the drone detection task may also be referred to as the classification task, which involves recognizing the received RF signals as either drone signatures or interferences [62,69]. Moreover, the drone make/model detection task, is also defined as classification task [56]. In general, drone detection is a two-step sequential process that includes detection and identification. More specifically, the detection step is the recognition of the presence of any object, which may or may not be a drone, and then determining whether the detected object is a drone, which is an identification step. After that, identification of drone type (make/model) is the drone classification. In the following, we discuss the modalities and procedures separately for RF and WiFi techniques related to the information collected from the surveyed literature.

4.1.1. RF-based

In this paragraph we discuss the approaches, methods and algorithms summarized in Table 5 in order to address RQ2 from what concerns RF techniques.

Detection via drone physical patterns. Detection of drones based on signals derived from drone communication and drone propeller was implemented experimentally in [43]. To illustrate the feasibility of the approach based on the reflection of the drone propeller signal, a single tone RF signal was transmitted to the drone and the reflected signal's strength was analyzed in both the time and frequency domains. The experiment was carried out using the Wireless Research Platform board. The changes in the signal due to propeller movement were observed graphically, concluding that such signal changes might be used to design a drone detection system. This method is not free of challenges: the size of propellers, the correct identification of drone propeller signal frequency with respect to other RF signal frequencies, and the quick detection of the drone due to its fast speed are major issues to be addressed while designing such a system. In this connection, [45] proposes to detect the presence of a drone by exploiting the variations in the wireless signal due to its body vibration and body shifting patterns. Wavelet Transform (WT) and Fourier analysis were used for body movement and body vibration, respectively. The detection of drones was confirmed by calculating the standard deviation of the received signal and then comparing it with the standard deviation of the environment.

It is worth noting that wireless signals originating from the drone are subject to variations due to the UAV's body shifting and vibration during flight operation. This aspect was also experimentally proved by [46] in terms of NLoS and LoS conditions. Received Signal Strength (RSS) measurements were taken in these two conditions and results showed significant differences. Although such conclusion is derived from an empirical test, it created the need to solve another issue: the system's ability to differentiate between LoS and NLoS signals. Therefore, the same research trend of analyzing the changes on RSS due to drone motion and vibrations, [46] provided significant contribution by designing a detection system based on NLoS signal, obtained after differentiating between NLoS and LoS components. Initially, the system classified the received RF signal as LoS or NLoS using a Supervised Support Vector Machine (S-SVM) learning model without any a-priori knowledge on the drone's presence. Once NLoS samples were identified, the drone's unique signatures due to the drone body's motion and vibration were extracted over the NLoS conditions by implementing Long Short-Term Memory Neural Network (LSTM-NN) to detect drones.

Detection via drone communication. Experimental work is done by [43] to eavesdrop on the communication signals of a drone in the category of passive approach and used the Software-Defined Radio (SDR) platform for this purpose. Experiments were carried out in a city and in an open field with additional WiFi-enabled devices, with a maximum observed distance of 50 m. In addition, according to [44,64], detection of a drone can be performed by exploiting the video streaming downlink signals sent from the UAV to the controller, which are based on Orthogonal Frequency-Division Multiplexing (OFDM) WiFi technology. Since OFDM uses Fast Fourier Transform (FFT) and Inverse FFT to perform multi-carrier modulation and demodulation, and a WiFi-based signal is generated frame by frame. The study of this frame using the Cyclic Autocorrelation Function (CAF) or the Spectral Density Correlation (SCD) yields cyclostationarity signatures. Based upon such principle, in [44] a table comprising the list of drones, their signal bandwidth and carrier frequencies was created in order to identify the different drones operating in the OFDM WiFi domain. The authors in [60] also employed OFDM signals transmitted from UAVs, with a particular emphasis on the UAV Flight Control Signal (FCS) and UAV Image Transmission Signal (ITS). The Field-Programmable Gate Array (FPGA) hardware is employed to implement and expedite the proposed detection algorithm.

When the drone surveillance system requires not only to detect a drone, but also to identify its type, model and flight mode, a fundamental building block consists in a-priori information, namely a huge database containing all relevant data about the potential UAVs to be recognized. Towards the realization of such a system, [47] proposed a system for building up the database storing such information. RF data was collected and analyzed systematically and different flight modes were employed, such as *off state*, *on state*,

flying, hovering, and video recording. Once the database was developed, Deep Neural Networks (DNN) learning approaches were designed to validate the system to detect and identify intruder drones. Concerning scenarios where also authorized drone can access a specific area, an approach for the detection of illegal UAVs is proposed in [51], where authors simulated a surveillance system where the signal transmitted from the drone is modeled as Independent and Identical Distribution (IID) with zero mean and unit variance. By contrast, the received energy signal is approximated as a Gaussian random variable when the number of samples is relatively large. Detection of an illegal drone is based upon the spectrum resource management unit. Detection of authorized drones only using both ground and aerial stations is presented in [79]. From this perspective, if an authorized drone is not working in the region, then an illegal drone must be present.

Authors in [52,63] also captured UAV communication signals. More specifically, [52] classified UAV signals from non-UAV signals via the BackPropagation Neural Network

(BPNN), whereas [63] utilized an un-supervised learning approach to detect intruders. To reduce the probability of missed detection, in [40] acoustic and video techniques were implemented jointly with the RF-based approach. A spectrum analyzer was used to extract drone communication features for the RF technique. All these three methods are processed in parallel, and then results are fused using the OR operator to minimize missed detection. However, given the purpose of the present survey, in this work only the results related to the RF-based technique were considered to the purpose of addressing RQ3 (see Section 5). Due to their mobility and versatility, UAVs become difficult to detect. By taking advantage of the enemy's Ground-Station (GS)-to-UAV transmission, a ground vehicle is used as a surveillance station to identify the hostile UAV in [59] by presenting numerical analysis. While dedicated RF sensor networks can detect drones in a limited area, they cannot provide wide-area coverage. Rather than employing specialized and expensive hardware for the purpose of detecting drones in densely populated urban regions with significant signal-to-interference level, the proposed solution in [61] utilizes terrestrial RF facilities that operate at the same frequency as the target drone. To optimize the drone detection across the network, the optimal parameters are determined, which include beam tilt angles and sensor density.

Autonomous drone detection. It is straightforward that both remotely operated and autonomous drones can be used for any malicious attacks. In case UAVs are autonomous, the drone does not send commands to its controller as it flies following a pre-defined or automatically determined route. Obviously, drone detection systems cannot exploit any communication signals between UAV and controller in order to recognize the presence of a potential malicious target. To this aim, [44] proposed a method based upon analyzing the micro-Doppler signatures on the spectrogram derived from third-party RF signals, thus adopting a passive RF technique. In detail, the system collects electromagnetic scattering and reflection signals produced by drones on third party opportunistic RF sources.

Detection via drone controller. The target of drone surveillance systems is not necessarily the drone itself, but it can also be the device controller. In such case, the captured RF signals are originated from the drone controller rather than the UAV itself. In this connection, the work presented by [21] was successively extended in [41] by taking into account also different interference signals, such as background noise, other WiFi and Bluetooth operating devices. Once the RF signal was captured, the UAV controller signal was identified by analyzing its bandwidth and modulation features. The rationale is to identify and classify the different UAV controllers based on their waveform transients, which are proven to be specific for different UAV controllers. A two-stage detector system was installed, in which one was used to decide that captured data contains an RF signal or not, and it was implemented using the two-state Markov model. After detecting the RF signal, the second detector identified whether the signal belongs to a UAV controller or any other interfering source. In case a UAV controller is detected, ML algorithms are used for feature extraction and identification of manufacturer and model of the controller. On the contrary, if the captured signal is a background interference, the system identifies whether the source is operating in WiFi or Bluetooth technology.

Furthermore, in [55,67], authors classified 15 different UAVs by capturing RF signals from their controllers. More specifically, [56] employed the Empirical Wavelet Transform (EWT) and CNN models for the captured signal decomposition and classification tasks, respectively. However, the system

proposed in [67] focused on lower values of SNR and obtained more efficient classification results. Similarly to [55], but with an emphasis on building a drone RF fingerprinting dataset, [65] recognized and classified UAVs using from their controller signals. Authors in [21,41], and [55] captured data from drone controllers, however as per [56], in real-world settings, drone controllers may be located far from the UAV's flight path, making it difficult to capture controller data for analysis.

Secure/covert transmission. Any malicious operator could operate a UAV employing covert transmission to cause harm to military or civilian targets. In order to tackle this issue, [50] presented an approach to detect when a UAV in covert communication mode spying over an unauthorized area. To this aim, an antenna array with beam sweeping has been used for monitoring the optimal number of sectors for the accurate detection of the drone. Channel State Information (CSI) between the surveillance station and the malicious UAV is obtained for the implementation of the beam sweeping scheme. One of the most employed techniques used for securing the communication between two devices is Frequency Hopping Spread Spectrum (FHSS), which makes the communication signals secure from interference and eavesdropping. In this perspective, [53] considered a scenario where the drone controller is transmitting signals to the UAV, securing communication through FHSS-Gaussian Frequency Shift Keying (FHSS-GFSK), and the drone is assumed non-cooperative. In this scenario, a system is designed to detect the FHSS-GFSK signals and recover the transmitted data bits using the Soft Detection & Compressive Sampling-Reduced Spectral Information (SD&CS-RSI) approach. Similar to [53,58] also targeted drone FHSS signals and differentiated them from other signals within the 2.4 GHz ISM band, by utilizing the cyclostationarity analysis. Nevertheless, in situations characterized by significant interference or crowded signal environments, achieving precise differentiation may provide a formidable challenge.

Table 5

Summary of Design Issues, methodology, and techniques employed in RF-based drone detection systems.

Ref.	Design issues	Methods	Techniques/ algorithm	ML/DL algorithms
[24]	Detection of drones with low transmit powers. Detection using surveillance drones Impact of SDr altitude on the coverage area.	Based upon RF signal power measurement	–	CNNs & LSTM
[40]	To distinguish drone RF signal from similar RF signals. To suitably deploy in Urban areas.	Eavesdropping drone communication signals.	USRP RF sensor & Spectrum analyzer	SVM Classifier
[21]	To address and overcome time-domain techniques issues in RF fingerprints. To detect targets at low SNR values. To detect UAVs via their controllers.	Capturing the uplink drone communication signals	Wavelet transform detector Markov models	NCA (Features extraction) kNN, DA, SVM and NN classifiers.
[41]	To detect & classify drone controller signals in the presence of interferences. To use energy transients in the RF feature extraction for effective classification.	Capturing the uplink drone communication signals	Wavelet transform detector Fourier analysis Markov models	NCA kNN, RandF, DA, SVM and NN classifiers
[42]	To eliminate the clutters. To detect drones with high accuracy. To detect multiple drones.	UAV communication signals and frequency estimation approach.	Spectrum accumulation and statistical fingerprint analysis	–
[43]	To receive RF signals over distance and denoise them. To detect drones in the presence of interferences.	Eavesdropping drone communication signals and drone propeller signals acquisition	Time-Frequency analysis	–
[44]	To detect autonomous drones. To select & implement the suitable hardware.	i. Cyclostationarity signatures from drone communication and Pseudo-Doppler principle based AoA estimation. ii. Acquisition & analysis of micro-doppler signatures. Drone body vibration and body shifting signals acquisition.	i. Cyclostationarity feature detection. ii. Micro Doppler feature estimation & Time-Frequency analysis.	–
[45]	To identify the drone unique signatures. To capture and validate the drone physical characteristics. To mitigate the noise interferences.	Drone body vibration and body shifting signals acquisition.	Wavelet transform detector and Fourier analysis	–
[46]	To differentiate the received signal into LoS & NLoS signals. To mitigate the multi path effect. To design cost effective system.	Drone motion and vibration patterns.	–	S-SVM learning model. LSTM-NN model.
[47]	To build a large database for different drones RF signal and their flight modes. To intelligently detect drone, its type & its flight mode.	Eavesdropping the UAV communication signals	Discrete Fourier Transform (DFT)	DNN Model
[48]	To analyze the drone control signals. To decode received signals for device identification & information interpretation.	Capturing the uplink drone communication signals	Demodulation techniques and Time-Frequency analysis.	–
[49]	To collect the wireless signals and build a database for different type of drones. To find a suitable model to classify the signals.	Eavesdropping the drone communication signals.	–	PCA for dimension reduction. AC-WGAN classifier
[50]	To detect UAV when its in covert communication.	Antenna array with beam sweeping scheme.	Likelihood ratio test	–

(continued on next page)

Table 5 (continued).

Ref.	Design issues	Methods	Techniques/ algorithm	ML/DL algorithms
[51]	To detect illegal drone in occupied spectrum.	Capturing & conversation UAV signals into an image.	Energy detector	VGGNet and ShuffleNetV2 DL models.
[52]	To select & modify the features from received signal for detection.	Eavesdropping the UAV communication signals.	Empirical Mode Decomposition (EMD)	Backpropagation NN model.
[53]	To use appropriate ANN model. To detect the FHSS-GFSK signals & to retrieve data bit sequences. To calculate the RF feature vector independent of the received signal.	Eavesdropping the uplink drone communication signals.	Soft Detection & Compressive Sampling-Reduced Spectral Information (SD&CS-RSI)	–
[54]	To identify, using spectrum data, the UAV signals. To detect UAVs by their signal frequency hopping.	Monitoring frequency hopping signals of UAVs.	Noise threshold calculation.	Improved K-means clustering algorithm.
[55]	To detect & classify different UAVs.	Capturing drone controller signals.	EWT	Deep CNN model.
[56]	To detect a specific UAV model. To analyze the non-overlapping downlink channel.	Capturing UAVs downlink RF signals.	Preamble feature extraction	CNN model, Federated learning
[57]	To analyze RF signal headers. To design a combined UAV detection and classification system	Exploiting spectrogram images of RF signals	–	YOLO (You Only Look Once)
[58]	To detect drone controller signals.	By exploiting FHSS communication signals.	Cyclostationarity feature detection, Short-term FT Monte-Carlo simulations	–
[59]	To detect a mobile UAV.	By exploiting GS-UAV transmission.	–	–
[60]	To develop algorithm to detect and identify UAV signals. To use FPGA for real-time processing.	By exploiting UAV transmission signals.	Time-Frequency analysis, FPGA algorithm.	–
[61]	To leverage existing terrestrial RF networks for drone detection. To improve drone detection in dense urban areas.	By exploiting non-dedicated RF sensing network.	Stochastic geometry tools	–
[62]	To classify UAV signals from interferences. To classify UAV signals, types and flight modes.	Eavesdropping the UAV communication and control signals.	HHT and wavelet packet transform.	SDAE, LOF, and XGBoost classifier
[63]	To use un-supervised learning to detect intruders. To achieve timely intruder detection.	Capturing uplink or downlink signals.	PSD, and spectrogram images analysis	k-means clustering, PCA, CNN model.
[64]	To detect UAV when UAV video signal and WiFi have identical bandwidth.	Capturing UAV transmission and WiFi signals.	FFT, spectrum analyzer.	ANN, SVM, DT, and RandF classifiers
[65]	To reliably recognize the UAV heterogeneity. To develop UAV RF fingerprinting based dataset.	Collecting UAV telemetry signals.	–	CNN models (VGG16, ResNet, and CLDNN).
[66]	To detect drones, and classify their types and operating modes.	Intercepting drone communication signals.	–	CNN model
[67]	To detect drones with low SNR values.	By capturing drone controller signals.	Signal thresholding and spectrogram transformation	CNN
[68]	To detect and identify UAVs.	By capturing wireless signals and extracting AIB, SIB, and FD features.	Haar wavelet transform, Thresholding, and Higuchi's algorithm.	PCA, NCA, SVM, kNN and RandF
[69]	To design a highly accurate UAV detection system.	By capturing the communication and CSI data.	EMD, FT, WT, and STFT	SVM, RF, NB, EL and KNN
[70]	To detect drones and their types.	Eavesdropping the UAV communication signals	FFT	CNN model

Multiple drone detection. Detecting multiple drones is also a challenging aspect of drone detection systems. The system designed by [42] was not only able to cover larger ranges but also to detect multiple drones. Drone communication signals were captured, and frequency estimation techniques were used to perform drone detection. Two UAVs were placed at the distance of 2400m and 2500m, and frequency estimation algorithms, Spectrum Accumulation (SA), and Statistical Fingerprint Analysis (SFA) were implemented to perform the detection of both drones simultaneously. Furthermore, with the aid of an improved clustering algorithm and adaptive noise threshold utilizing frequency hopping signals from UAVs, the authors in [54] have developed a system that is capable of recognizing multiple drones, two in the paper's test case scenario. Since multiple UAVs in the same spectrum may use FHSS, complicating signal isolation from various sources. Therefore, [56] designed a UAV classifier based on the downlink, as the involved UAVs possess distinct and non-overlapping downlink channels. Drone detection in [56] is based on receiving the IQ (In-phase and Quadrature) raw data. However, according to [57], spectrogram representation provides more information than conventional representations such as Power Spectral Density (PSD) or IQ representations of the signal.

Aerial-based detection. In parallel with implementing terrestrial detection systems, aerial stations are also under investigation to accurately detect amateur drones. [24] has designed an aerial surveillance system for the detection and localization of amateur drones and showed that the coverage area is enhanced if the surveillance drone is placed at optimum altitude.

Drone classification. Jointly with the detection of drones, studies in the literature also cover classifying the specific type of drone. For instance, [21,41,45,47,49,62,66] propose specific solutions to determine the type of UAV involved in the detection process. In [49], some UAVs were operated in a quiet indoor environment to collect wireless signals to build a database. Hence, Wasserstein Generative Adversarial Networks (WGANs) was selected as a suitable model for the identification and classification of UAVs, since it required a smaller number of samples to train a model compared with other available techniques, such as k-nearest neighbors (kNN). Since the collected wireless data had high dimensions, authors reduced them by employing Principal Component Analysis (PCA). In the experimental setup, four UAVs were deployed, namely Phantom, Hubsan, Mi, and Xiro, all exhibiting wireless signals different from each other. Similarly, [41] described methodology of classification after the detection of drone controller signal. So, once the UAV controller signal is identified, energy transients are extracted to obtain the drone's features, which are further processed by a Neighborhood Component Analysis (NCA) for feature minimization. Hence, machine learning algorithms are used to classify the UAVs. In this perspective, five different ML classifiers were tested: kNN, RandF, Discriminant Analysis (DA), SVM, and NN. Monte Carlo simulations are performed for the analysis of classification results both in [21,41].

Remote identification (Remote ID). A recent standard introduced by FAA in [80], regarding the identification and accountability of UAVs, enables drones to broadcast their location and identification information, and thus allowing governing bodies, law enforcement, and the general public to identify drones remotely. The following are the primary reasons to introduce such a standard. (i) *safety*: It improves safety by enabling both governmental and civil identification of UAVs and thus helps to avoid collisions, and airspace violations. (ii) *security*: It aids in recognizing legitimate UAVs from unauthorized or potentially harmful ones, and thus could help authorities to respond to security threats quickly. (iii) *compliance*: The standard defines performance requirements for Remote ID to ensure that UAV operators follow rules. (iii) *privacy*: In addition to UAV identification, the standard seeks to protect the operational privacy of remote pilots. In [81], authors conducted a survey and highlighted the activities carried out in regulating, standardizing, designing, deploying, and testing the remote ID.

The remote ID, which transmits sensitive information like location and identification, raises privacy and security issues among UAV operators. Furthermore, while Remote ID can detect authorized and un-authorized drones, it can also expose critical information to malicious parties. The coordination between legal drones and counter-drone systems is challenging in real time. From this perspective, [82] presented a set of protocols as a solution to this challenge. Furthermore, [83,84] addressed the privacy concerns of UAV operators and the security concerns of legal UAVs from malicious attacks, respectively, while implementing the remote ID standard. The weaknesses of remote ID, strategies used to disrupt safety and accountability, and solutions to address privacy concerns are outlined in [85].

4.1.2. Techniques & algorithms included in RF category

Apart from experimental hardware setup, different tools and techniques are included in all drone detection system stages. A list of sensing techniques, feature extraction algorithms, and ML classifiers used for RF-based drone detection are mentioned in Table 5. Different techniques and algorithms have been adapted for specific tasks at different stages of the detection system. For instance, the framework proposed in [62] implemented machine learning techniques, such as the Stacked Denoising Autoencoder (SDAE) and Local Outlier factor (LOF) to denoise the data and detect outliers in the dataset, respectively. Moreover, feature extraction makes use of signal decomposition methods such as the Hilbert-Huang transform (HHT) and the wavelet packet transform. Besides, in [21,41], energy transients extracted from the captured RF signals in the energy-time-frequency domain using the spectrogram method, from which transmitted frequency and frequency hopping patterns of received RF signals are revealed, indicating detection information. RF fingerprints information was extracted from energy transients, and the extracted features were skewness, variance, energy spectral entropy, and kurtosis. Finally, these statistical features were used to train the ML classifiers kNN, DA, SVM, and NN to infer the type of drone. In [21,41], NCA is utilized for feature selection and minimization from extracted fingerprints, whilst PCA is used in [49] for dimensionality reduction. In contrast to [41], which employed energy transients as RF fingerprinting signatures, [68] demonstrated improved UAV identification performance by extracting Axially Integrated Bispectra (AIB), Square Integrated Bispectra (SIB), and fractal dimension (FD) features as RF fingerprints. Similarly, [69] applied FT, WT, and Short Time FT (STFT) to obtain the Signal Frequency Spectrum (SFS), Wavelet Energy Entropy (WEE), and Power Spectral Entropy (PSE) features,

Table 6

Summary of Design Issues, methodology, and techniques employed in the WiFi-based drone detection system.

Ref.	Design issues	Methods	Techniques/ algorithm	ML/DL algorithms
[15]	To detect privacy invasion attack.	Based upon an attacking model.	Standard deviation	–
	To design detection system for residential areas with cheap COTS hardware.	By capturing the packets of UAV with video transmission.		
[22]	To detect unauthorized drones.	WiFi statistical finger print analysis.	WiFi sniffer	Logic Boost, Random forest, Random tree, and SMO classifiers.
[71]	To identify near by drone threats.	WiFi statistical fingerprint analysis including video streaming flow.	WiFi sniffer	RF, J48, and NB classifiers.
	Accurate & reliable drone detection irrespective of live video transmission.			
[72]	To generate fake statistical distribution of WiFi features.	WiFi fingerprint analysis during fake WiFi traffic	Wildcard PDF Estimation (fake traffic)	Random Forest Tree (RFT) model
	To detect the drone during fake WiFi packets emission.			
[73]	To quickly detect UAV over encrypted WiFi traffic	Analysis of features extracted from Encrypted WiFi traffic.	WiFi sniffer	One vs all logistic regression multi-class classification algorithm with re-weighted ℓ_1 -norm technique.
[74]	To quickly detect UAV over encrypted WiFi traffic.	Analysis of features extracted from Encrypted WiFi traffic.	WiFi sniffer	One vs all logistic regression multi-class classification algorithm with re-weighted ℓ_1 -norm technique.
	To identify UAV's operation mode.			SVM and RF classifier.
[75]	To identify the state of the drone movement over encrypted WiFi network.	Eavesdropping the encrypted WiFi based drone communication	wireless sniffer	Trees-J48, NNs, and Random Forest.
[76]	To make cost effective design.	Eavesdropping the encrypted WiFi based drone communication	Wireless sniffer	Random Forest classifier.
	To identify the drone, its current state & to distinguish its movements.			
	To analyze the detection delay and effect of packet loss.			
[77]	To design cost-effective detection system for urban scenarios.	Mobile crowd-sensing approach.	WiFi signals sensing via Smart devices.	–
	To distinguish drone WiFi signals from other WiFi APs.	Analysis of WiFi beacon information and drone MAC addresses		
	To attract participants for sensing task.			
[78]	To identify a spying drone.	Adding watermark to the encrypted video data by applying a periodic physical stimulus	WiFi signals sensing, FFT, and a classification algorithm	–

respectively, to detect UAVs. Indeed, selecting and implementing the most appropriate classification algorithm plays a crucial role in achieving satisfactory detection and identification performance. However, choosing the most suitable classification algorithm is not the only posed challenge, since the choice of the feature extraction and selection technique is also a fundamental step, of paramount importance in obtaining the optimal system performance.

Another type of approach was proposed in [44], where time-frequency analysis was carried out using spectrogram plotting for micro-Doppler signatures to detect the autonomous drones. The reason for adopting this approach was the non-stationary and complicated nature of micro-Doppler signatures. Furthermore, background filtering and linear trend suppression were used to minimize the static and non-static cluttering from the received RF signals in [42], and finally SA and SFA techniques were adopted to confirm the presence of the UAV. For the detection of WiFi signals, Bandwidth computation using FFT analysis was implemented and Zero-crossing GFSK/FSK demodulator was used along with FFT analysis to distinguish the Bluetooth signals in [41].

4.1.3. WiFi-based

Detection and identification of drones using a WiFi-based approach are based on the rationale that communication between the drone and its controller can be eavesdropped, and features can be extracted from wireless traffic at the data link layer. to address RQ2.

Detection via capturing WiFi traffic. Commercial off-the-shelf (COTS) hardware, namely a WiFi receiver consisting in a Raspberry Pi along with a Wi-Pi USB WiFi adaptor, is used to deploy the detection system designed by [15] working in LoS scenarios. WiFi packets were captured in monitor mode and separated based upon Media Access Control (MAC) addresses. The system design is based on an attack scenario in which three cases were considered: launching, surveillance, and escaping of a drone. Although this technique proves effective, the considered scenario not always holds when a drone attacks. Indeed, drones may not be in proximity of the target and in LoS condition as discussed in [46]. In addition, the research in [78] aims to determine whether a passing drone is being used to spy by focusing the cases in which the victim is either indoors or driving. Furthermore, WiFi statistical fingerprint-based drone detection system is proposed in [22,71]. Since drones not only send control signals to their controllers but also live video feeds, this latter streaming component might be preponderant with respect to control traffic, thus introducing peculiar features in the drone-controller communication. Hence, communication signals to and from the drone excluding video transmissions could deceive a traditional surveillance system and threaten with possible malicious attacks [71]. Therefore, advanced anti-drone systems are required to be able to detect the presence of unauthorized drones independently of whether the drone is transmitting video feed or not. Thus, [71] designed the detection system in which the detection of drones was analyzed both in the presence and absence of video transmission. Special focus is given to the stealth configuration in which the video of the drone is turned off. ML algorithms are utilized to classify and recognize the different classes of captured traffic.

Detection via crowdsensing. The literature also offers works exploiting crowdsensing for drone detection. Since the concept of crowdsensing consists in involving individual users to collaborate at sensing the surroundings using their devices, [77] proposes to perform the task of UAV detection by considering the sensing data collected from multiple individual users. MAC addresses and WiFi beacon information is used in designing the detection algorithm. To this aim, drone produced by different manufacturer are included in a comprehensive database, built with the help of participants. Furthermore, based on historical data, MAC addresses of static WiFi Access Points (APs) are removed and mobile APs (i.e., cellular phones, tablets, mobile hot-spots, etc.), are also excluded based on a cooperative decision made by all the sensing nodes. The information on newly recognized drones are incrementally included in the database. In this work authors claim that the designed system can also be employed to detect multiple drones simultaneously.

Detection in fake/encrypted Tx. To create a clear road map for effective, smart, and intelligent WiFi-based surveillance systems, [72] analyzed the performance of the system designed in [71] when drones deliberately deceive the guardian system. Therefore, [72] adopted an adversarial approach, by introducing the concept of fake WiFi traffic being transmitted by the intruder when it enters an unauthorized area in order to deceive the surveillance system. In order to generate the fake WiFi traffic, a detailed analysis of the statistical features of the WiFi communication was carried out first using the Distribution fitting toolbox in MATLAB software considering both drone and background transmissions. Based upon the estimated Probability Density Functions (PDF) of drone and background traffic, a normal PDF function centered at an intermediate point of these two estimates is generated to create a wildcard PDF. Features used in the generation of fake data were: total number of packets, duration of the traffic flow, average packet length, Root Mean Square (RMS) of packet length, average and RMS of Inter-Arrival Time, number of embedded packets, and number of DS-Mode packets. As expected by the authors' claim, system performance for detecting the intruder with such fake data was reduced to 50%, when considering the same system proposed in [71] and using the Random Forest Tree classifier.

With the advancement in designing the drone surveillance system, there is also advancement in technologies and techniques being used by malicious users to fly their drones to deceive monitoring systems in order to perform malicious tasks. Some examples are given by the following situations: a drone can be operated by using an open-source operating system, where the remote control operator can dynamically change the MAC addresses of both transmitter and receiver; the drone can be operated in GPS denied area, or its GPS can be switched off; communication between the drone and the controller can be encrypted, which prevents the packet inspection by the surveillance system. In addition, frequencies of drone communication signals cannot be jammed in case these are also being used by other devices and applications in the unauthorized zone.

Towards the task of detection for the encrypted traffic, [73,74] designed a drone surveillance system to detect and identify drones. In these works authors especially focus on the fast detection and identification of the drones. A design matrix, containing the function for computing the statistical features, was created for each dataset. A total of 12 features were computed, indicated along with their computational time. Hence, a re-weighted ℓ_1 -norm based cost function was formulated to eliminate the features which convey no information, and subsequently another selection was performed on features contributing equally in UAV identification, based upon their computational time. To achieve this, a one-vs-all logistic regression function in the category of multi-class classification was implemented.

In this connection, [75,76] also proposed a drone detection system devised to identify targets with encrypted WiFi transmissions. In [76], the proposed system was designed to perform multiple tasks, including the detection of the drone and identification of its current state (steady or moving), determination of the drone movement, analysis of the delay for quick detection, and the effects of packet loss on system performance. Initially, packet size was investigated in detail for all cases in which both drone and controller were sending and receiving information to each other regarding the steady and moving status. Hence, inter-arrival time patterns were analyzed for differentiating events in the same flow. In addition to this, Information Gain Maximization approach was used to create a decision tree for classification. Furthermore, a weighted approach was also used, as adopted in [74], to investigate the contribution of each feature in the detection task, towards optimizing the performance in terms of detection delay.

Table 7
Summary of Design Issues, methodology, and techniques employed in the RF-/WiFi-based drone tracking system.

Ref.	Design issues	Methods	Techniques/ Algorithm	ML/DL Algorithms
[34]	Direction & location finding of drone.	DoA based localization	Triangulation	–
	Tracking the intruder.			
[35]	Tracking of moving aerial RF source.	RSS position estimation	FIM optimization & RH approach	–
[36]	Tracking of intermittent UAV.	RSS position estimation	Steepest descent posterior CRLB	–
[37]	To examine EKF-based tracking performance when using different motion models.	TDoA based localization	EKF, geolocation software N6854A.	–
[38]	To track UAVs without relying on active communication from them. To use cellular connectivity for UAV tracking.	Using asynchronous measurements of Doppler shift from base stations.	IMM Filtering, EKF	–
[39]	To track unauthorized drones.	RSSI position estimation	CDQA and ADCA algorithms.	–

Drone movement identification. To determine the drone activity, in [76] seven different types of drone movements were considered and detected, namely pitch down, pitch up, roll left, roll right, throttle up, yaw left, and yaw right. To collect the samples for these movements, drones' propeller was un-screwed, up to 1-minute samples were gathered, and they were sent to the classifier for their identification after processing. Moreover, detection system performance was evaluated when experiencing packets loss and in the presence of evasion strategies, proving that these two factors significantly affect the detection performance, since the detection rate strictly depends on packets inter-arrival time. When considering an outdoor scenario at long distances with the presence of obstacles, packets loss was observed up to 74%. However, the performance of the detection systems was still observed satisfactory, reflecting the system robustness [76]. Furthermore, authors analyzed an evasion technique consisting in inserting ad-hoc delay in packet transmission, thus leading to an alteration of packet inter-arrival time. This delay time was different for the different indoor and outdoor scenarios, however the maximum observed time was 80 ms, even though this parameter also depends upon the latency, drone response time, and maneuverability.

Decoding of operation mode. It is normally assumed and observed that the drone stands hovering in a static place while sent to an unauthorized area for any task like spying, privacy invasion, and the status of a drone operation to be performed is received from its controller. Some scenarios must decode the status of the operation the drone is performing. To this aim, [74] tried to identify the status of the operating mode of the drone along with the detection and identification of the UAV type. SVM and random forest multi-class classifiers were implemented to decode the operation mode with the incoming traffic flow as input and identification of the operation mode of the detected UAV as output. Also in this case, authors identify the drone status among a predefined set of UAV operating modes.

Covering long ranges. In [86], the authors proposed the Internet of Drones (IoD) architecture for surveillance applications that utilize IEEE 802.11ah standards. This standard, which works in the 1 GHz frequency band and provides energy-efficient and long-range communication, can be used as a ground station with 802.11ah access points (APs) to detect drones up to 1 km away with a single AP. The AP can gather and evaluate the drones' RSSI information to monitor them. To extend coverage beyond 1 km, an additional AP or drone can be used as a relay node and aerial surveillance station alongside the original. Nevertheless, an additional AP or drone acting as an aerial surveillance station and relay node can complement the first one and extend the coverage area beyond 1 km.

Drone classification. Among the surveyed works, the classification of the type of drones with a WiFi based approach is not as studied as the detection task. However, [22] used traffic of another flying drone along with the traffic of WiFi AP and then implemented ML classifiers to classify the drone traffic with other terrestrial drones. On the same line, [73,74] identified four and eight different types of drones, respectively, by using multi-class classification algorithms. The considered types of drones are summarized in Table 9.

4.1.4. Techniques & algorithms included in WiFi category

A list of techniques, tools, algorithms, and ML classifiers employed to devise the reviewed WiFi-based drone detection systems is presented in Table 6. Wireshark is employed as a sensing tool for capturing the WiFi traffic, while ML-based algorithms have been used for drone detection and classification based upon some useful features extracted from the captured data. For example, in [22,71], fundamental features extracted from the captured WiFi traffic are: total number of packets, total duration of the flow,

average packet length, root mean square of packet length, average packets inter-arrival time, RMS of packets inter-arrival time, number of embedded packets and number of packets with specific DS status flags. Among the extracted features, *embedded packets* were used as the indication of a possible WiFi network card. However, in [71], an information gain-based attribute selection algorithm was applied to reduce the set of fingerprint features from 11 to 8, and this reduced feature set was used to perform drone detection in stealth scenarios.

4.2. Drone tracking part

4.2.1. RF-/WiFi- based

The review carried out for articles related to drone tracking approaches highlighted that the main techniques used for the tracking task is currently based on exploiting the RF signal. In Table 7 we summarize the methods and procedures proposed in the considered literature studies. Concerning systems involving a ground-based surveillance station, [34] performed the task of tracking by locating the intruder first. Although the main issue tackled in [34] was the localization of the unauthorized drone and its controller, authors also propose an approach to track the UAV, based upon the data collected for localization. Towards finding out the direction and location of the drones, [34] used the triangulation approach based upon the Angle of Arrival (AoA) technique. To this aim, a directional antenna is pointed as 0° north, and the angle of the drone is identified as the angle where cross-correlation of the signal is the highest. The authors used two sensing stations at the receiver yielding two AoA, which encouraged the utilization of the angle-side-angle solution to find out the drone's location and its controller.

Concerning the tracking task using aerial monitoring stations, [35,36] proposed a scenario in which the moving RF source is tracked by a UAV swarm. Focus is given on path planning for the UAV swarm, by exploiting the target location estimation technique presented in [35]: at each time instant, the state vector of the estimated target is updated by taking the information of the target position and measurements of the tracking model. Hence, by optimizing and predicting the D-optimality criterion, Cramer–Rao Lower Bound (CRLB) is ensured on the localization error. In particular, authors exploit the estimated trajectory of the target UAV to determine the tracking path for the surveillance drones. Three tracking UAVs and one target UAV were considered with the assumption that RSS measurement from the target UAV is always available to the surveillance UAVs.

Sometimes intruders use intermittent signals in order to escape from the eye of the surveillance system. Therefore, one of the important aspects of the surveillance system in tracking the intermittent RF sources is reacquiring the target position when it stops the transmission before being fully localized and tracked by the surveillance system. To cope with such a scenario, two tasks were performed by [36] for tracking of the intermittent RF source using UAV swarm. One is to estimate its position, and the second is to plan the trajectory for the surveillance UAVs with the lowest localization error. Steepest descent posterior CRLB and bioinspired heuristic path planning approaches have been implemented. The steepest descent approach aims at maximizing the measurement information content and this information is represented by the Fisher Information Matrix (FIM). A stochastic process has been used to model the moving RF source and Received Signal Strength Indicator (RSSI) measurements. Further, a Bernoulli process is employed for the transmissions from the target, and the received power is modeled by taking into account the Friis channel attenuation, log-normal fading, and by adding thermal noise. D-optimality criteria and maximization of the FIM matrix is adopted for the assessment of the path planning approach. To this purpose, the direction towards which the UAVs shall move is determined by solving the path planning optimizing problem. The considered optimization problem also requires target estimation and distance information achieved through a Bayesian estimator and detection-based Extended Kalman Filter (EKF). In terms of D-optimality criteria, the presented results showed that the steepest descent approach has optimum performance compared to the bioinspired heuristic approach. Similar to [36,39] estimates the location (distance and angle) of the targeted drone based on the RSSI signals of four Bluetooth Low Energy (BLE) receivers. The system tracks the target progressively in accordance with the estimated distance and angle information using the Constant Distance and Quantized Angle (CDQA) and Adaptive Distance and Continuous Angle (ADCA) algorithms.

In [37], four N6841A RF sensors are used, separated 50 m apart, to gather IQ RF data. This data was then processed by geolocation software, which used the Time Difference of Arrival (TDoA) technique to determine the UAV's location. The three motion models (constant velocity, constant acceleration, and constant turn) are used to simulate the arbitrary UAV trajectory. Each segment of the UAV's trajectory will employ a different motion model and EKF adopts these motion models to improve the localization accuracy. Since TDoA approach requires network-wide time synchronization with high-accuracy, which is difficult and expensive. The cellular base stations, on the other hand, are frequently frequency synchronized, making Doppler shift measurements a technological advantage. The challenge of tracking UAVs without actively communicating with them is discussed in [38] by taking into account the asynchronous Doppler shift data from 4G and 5G base stations. The Interacting Multiple Model (IMM) filtering algorithm, proposed in this study, processes Doppler shift data and estimates 3-D velocity, position, and carrier frequency bias of the UAV.

4.2.2. Techniques & algorithms included in RF/WiFi category

Concerning the tracking of drones using UAV(s), the concept of tracking is implemented in terms of UAV path planning, and different techniques and algorithms have been used to perform this task, as summarized in Table 7. In this connection, [36] shows that the steepest descent posterior CRLB is more promising in maximizing the information through D-optimality criteria compared to the bio-inspired algorithm, but at the cost of higher computational complexity. Similarly, the Receding Horizon (RH) algorithm in [35] has produced more effective results for tracking in terms of estimation error with respect to the basic bio-inspired algorithm without prediction. In addition, [35,36] used a location estimation algorithm known as the EKF algorithm. However, EKF provides

good performance in terms of computational cost but at the same time it affects the minimization of the localization error, achieving lower performance with respect to recursive Bayesian estimation [36]. The algorithm designed in [38] leverage the concept of IMM filtering, where the nonlinear measurement equations are efficiently handled through the utilization of EKF. Furthermore, the sensitivity of the EKF to the fidelity of data alignment was emphasized in [37] where the EKF reduces estimation errors when the elements of the UAV trajectory are aligned with the motion models under consideration.

5. SLR RQ3 (Performance evaluation & limitations)

5.1. Drone detection part

Based upon the collected data presented in Tables 8 and 9 for RF and WiFi techniques respectively, RQ3 is addressed. Performance evaluation of the detection systems using RF and WiFi techniques is analyzed and discussed here, along with potential limitations and influencing parameters. Different metrics are used as evaluating parameters for the performance analysis of drone detection systems. Some of the main employed criteria are listed in the following:

- Employed detection method
- Detection rate/probability
- Detection accuracy vs. SNR, environment & distance.
- False alarm rate/detection error
- Working range vs. environment
- Computational complexity
- Classification accuracy
- Individual classifier output in terms of True Positive Rate (TPR) and precision.
- Cost Effectiveness

5.1.1. RF based systems

Parameters influencing detection output. The main parameters influencing the detection performance in RF-based systems are summarized in Table 8. In this connection, SNR value has a huge impact on the detection performance. Good detection performance requires a high value of SNR. For instance, optimum detection results are obtained in [41], achieving a detection rate up to 99%, yet at the cost of an SNR value of 20 dB. Similar results are observed in [21], ensuring a SNR value of 10 dB. The reviewed works highlight that environmental effects cannot be ignored in the outdoor scenario, indeed in [21,41] higher performance has been obtained in indoor environment even with lower SNR values. In particular, a drone classification accuracy of 98% can be achieved in [49] in indoor environment with an SNR value of only 5 dB. Moreover, it is important to notice that only 4 different types of drones were considered by [49], while 17 types of drone controllers were included in [41]. A larger set of possible classes to be identified also significantly affects the detection performance.

The performance of the detection systems discussed so far is related to remotely operated drones. However, [44] designed a system to detect also autonomous drones, where the detection was confirmed by plotting and analyzing time-frequency plots. Since micro-Doppler signatures are different for diverse types of drones, the implementation of such a system is quite challenging. In addition to this, the NLoS scenario also poses significant issues in achieving this goal. It has been observed and discussed that system outcome is influenced by the NLoS conditions. Therefore, in order to create a LoS with the targeted drone, [24] employed a surveillance drone placed at a high altitude, so that unauthorized drones can be detected easily from above. With this technique, intruder drones were detected even with very low values of SNR (*i.e.*, less than -15 dB) when considering an optimum altitude and a coverage radius of 500 m.

Detection of target drone in covert communication was discussed in [50], where numerical results were presented for minimum detection error probability versus the number of sectors, the elevation angle of UAV at the ground station, and the number of sectors with different values of noise power. Obtained results indicated the effectiveness of the proposed scheme towards RF-based detection of covert transmission. Nevertheless, a UAV with a single antenna was assumed in this study, which may not always be the case in real scenarios. Furthermore, towards detecting drones with secure transmissions, [53] assessed the performance of the proposed detection system in terms of Bit Error Rate (BER), which indicated that neither it is sensitive to the GFSK modulation index variation nor it changes for different sampling branches. However, the analysis of system performance for the noisy case was not considered in this work.

ML vs. Non-ML based system performance. Globally, ML- and CNN-based drone detection systems show better performance when compared to non-ML based systems [46,47,49,70]. On the other hand, non-ML based systems can also achieve more than satisfactory results, as proved by the system designed in [42], providing a detection rate of 90% up to a distance of 2800 m, and by the BPNN model-based system achieving 82% detection accuracy up to a distance of 3 km. It is important to notice that also the complexity of the hardware employed to design the surveillance system (and, consequently, its cost) has a huge impact on the final anti-drone system detection performance. As a consequence, researchers and manufacturer always aim at a reasonable trade-off between different influencing parameters and the best performance possible. Finally, it is interesting to observe that the trend of designing ML-based systems has increased drastically in the latest period.

Drone type classification. Classification accuracy of 98.4% was observed with SVM as optimum classifier for fourteen different UAV controllers and kNN with accuracy of 95.53% with seventeen different controllers in [21,41], respectively. More accurate results in [21,41] are at the cost of higher SNR values as compared to [49]. In this connection, as expected, higher SNR values provide better system performance. In addition, also the signal sampling rate and the computational time are important parameters that cannot be ignored while designing a drone surveillance system. The performance of classifiers can also be influenced by other data, such as the number of flows captured, the size of the data, the number of samples, and the timing window, which can affect the performance of the classifiers in terms of accuracy and true positive rate. It is a general observation that kNN and SVM based methods usually require a rather large number of samples to train the system model [49]. However, the DA classifier is generally known for the shortest computational time, yet the observed performance was not satisfactory compared with kNN and RandF [41].

5.1.2. WiFi based systems

A system aimed at recognizing the type of attack performed by the unauthorized drone was designed in [15], where the detection of a malicious drone was performed through graphical analysis of the captured signals, when the value of standard deviation exceeds a certain threshold. The launching distance range was kept between 55 m to 65 m. Different flying patterns were considered for system evaluation, including straight, zig-zag, back and forth, NLoS over the roof, and NLoS around the house. Authors assumed that an anti-drone system is installed at the window of a building, and the drone shall be closed enough to the window and in proximity to take a video recording of the interior of the building. In such scenario, the system achieved better detection performance when considering a LoS target, rather than NLoS.

Delay-aware techniques were employed in both [74,76], where quick and fast detections are considered a requirement. In this case, the detection time proves a fundamental parameter in the system design. In [74], the system was able to perform the drone detection and identification within 0.35 s, while in [76] a delay up to 0.28 s was observed. Furthermore, the system designed in [74] was devised to operate in any outdoor scenario, such as campus and urban areas, with a 70 m range in LoS and up to 40 m in NLoS. Nonetheless, the achieved performance may not be sufficient if a larger area needs to be covered. In addition to detection, [74] also identified eight different types of drones with an average accuracy of 85.7–95.2%. Anyway, this system requires prior information regarding drone patterns for their correct detection and identification. Two main constraints were observed in the proposed WiFi approach: drones must be remotely operated and systems can only detect up to the WiFi coverage range. On the contrary, larger monitoring areas can be covered both in LoS and NLoS conditions in [77], but the detection approach is related to the correct organization of the crowdsensing platform, requiring active involvement and efficient coordination between participant users. Another disadvantage is the capability of detecting WiFi-operated drones only.

5.2. Drone tracking part

The performance of the drone tracking system is evaluated, and possible limitations have been identified as reported in Table 10. Usually, the performance of designed systems is evaluated concerning the nature of their output and the way the systems are developed. Therefore, performance evaluation metrics could vary depending upon the nature of the deployed surveillance station. In the following we identify a reasonable set of criteria used to assess the performance of the reviewed anti-drone systems:

- Analysis of D-optimality criteria
- Trend of CRLB
- Estimation error with respect to time, altitude, and distance between surveillance drones
- Convergence of trajectories towards the target
- Computational time & complexity

In [35], the performance was analyzed concerning the error, which was minimized with respect to detection time as surveillance UAVs were approaching the target drone. It is relevant to notice that this work provides only results related to simulations, while a real-life implementation of the considered system might imply specific trade-offs related to costs, range, accuracy, precision, energy efficiency, portability, and complexity, based on the employed technology. In addition, the use of UAVs for tracking tasks is normally subject to multiple issues, such as the efficient coordination between the ground station and the UAVs, different architectures for UAV deployment and specific path planning and drone tracking schemes. Concerning anti-drone systems based on ground

surveillance station, experimental tests in an open outdoor scenario were carried out in [34], where the location of drones and their controller were identified based on graphical analysis of the signals, and tracking was performed through AoA estimation. However, experimental tests showed significant error affecting in the presence of obstacles and interference, producing a considerable impact on the tracking performance.

6. SLR RQ4 (Effectiveness & generalization)

6.1. Drone detection part

Tables 8 and 9 highlight that the large majority of the surveyed systems provide not only a simulative analysis, but also experimental tests. Multiple field experiments were carried out by utilizing different hardware, and a large part of the works considering RF-based systems employed SDR boards for their implementation. However, these boards are usually expensive, thus implying high cost for the system. On the other hand, some works also propose cost-effective hardware, such as Raspberry Pi

Table 8
Summary of deployment and performance analysis of RF-based drone detection system.

Ref.	Surveillance station	Experimental/ Simulation	Environment	Drones/ Controllers	Constraints	Range	Outcome
[24]	Aerial	Simulations	Urban	drone	Environment dependent. Valid to only tele-operated drones.	Coverage radius beyond 2000 m	Detection of drones beyond -15 dB with optimum altitudes.
[40]	Ground	Experimental	Outdoor	DJI Phantom 4, DJI Phantom 3 and a Mavic Pro.	Low detection rate at distance of 100 m. Collected drone signatures are not generalized for all type of drones.	100 m	Detection = 73.3% Error = 2.8%
[21]	Ground	Experimental	Indoor	14 different UAV controllers	No NLoS considerations. Designed framework is limited to controlled environment. Interferences were not considered.	130 m	At 10 dB SNR: Detection Accuracy = 84% Optimum classifier (SVM) Accuracy $\approx$ 96.84%
[41]	Ground	Experimental	Both indoor & outdoor.	17 different UAV controllers	Several hours required for training the model. Necessary to operate receiver beyond 15 dB for accurate UAV identification.	Not specified	At 20 dB SNR: Detection accuracy = 99.9%. Optimum classifiers (kNN) Accuracy = 95.53%
[42]	Ground	Experimental	Outdoor	Phantom 4 UAV, DJI	Cost effective & NLoS scenario analysis of the designed system are missing. UAV detection in the presence of similar frequency RF signals.	2800 m	One UAV: Detection rate = 90% (2800 m) Two UAVs: Detection error = 0.2%
[43]	Ground	Experimental	Urban area & open field	WiFi drone	Small gain of receiver. Short distance. Prior information of drone communication signal & propeller signal frequency.	50 m	Drone signature analysis based detection indication.
[44]	Ground	Experimental	Open Campus	DJI Phantom 3	Drone detection is dependent upon video transmission from drone. Micro-Doppler signatures are different for all drones & their structure. Difficult to implement in NLoS scenarios.	Not specified	Pictorial representation of drone detection via Time & Frequency plots.
[45]	Ground	Experimental	Urban, Sub urban and campus.	Protocol Dronium, Sky Viper, Swift Stream, Parrot Bebob, Parrot AR Drone, Protocol Galileo Stealth, and DJI Phantom	Can only detect the drones for which the physical patterns are known in prior. Drones with different body structures & physical characteristics. Can classify the multiple drones but cannot detect multiples.	Urban area: 100 m sub urban area: 600 m Inside the campus: 200 m	Accuracy (urban) = 89.4% Accuracy (sub-urban) = 84.9% Successful classification of 07 different types of drones.
[46]	Ground	Experimental	Residential	DJI Spark and a self-built drone	Detection is dependent upon the known patterns. Can only detect single drone.	WiFi range	Accuracy = 97.02%

(continued on next page)

Table 8 (continued).

Ref.	Surveillance station	Experimental/Simulation	Environment	Drones/Controllers	Constraints	Range	Outcome
[47]	Ground	Experimental	Indoor	Parrot Bebop, Parrot AR Drone and DJI Phantom 3	System performance decreases with increasing no. of classes. To have RF patterns of drone flight modes in prior. Labeling of huge RF database.	Not specified	Detection accuracy = 99.7% Detection & type identification accuracy = 84.5% Detection, type identification & flight mode accuracy = 46.8% Time-Freq: analysis based drone detection.
[48]	Ground	Experimental	Not specified	Optic 6, Futaba T7SG FASST, Graupner MX20, Jeti and DJI Phantom 3 controllers.	To have prior knowledge of the interpretation of data bit groups in the transmitted signals. The decoding process is dependent upon the specific device & received signal frequency.	Not specified	
[49]	Ground	Experimental	Both indoor & outdoor	Phantom, Hubsan, Mi & Xiro	Database is limited only for four different UAVs.	200 m	Classification accuracy (Indoor) = 98% Classification accuracy (Outdoor) = 62.5%–98%.
[50]	Ground	Numerical analysis	Air-to-Ground channel	Generalized	Single antenna was considered at UAV side. Valid for tele-operated drones only.	100 m	Numerical analysis of detection error probability (minimum).
[51]	Ground	Simulation	Low SNR environment	IID with zero mean and unit variance model	Valid to only tele-operated drones. Trade off between accuracy and computational complexity. Trade off between no. of sensors & classification performance.	1000 m × 1000 m square	Classification accuracy (VGGNet) = 83% Classification accuracy (ShuffleNetV2) = 92%.
[52]	Ground	Experimental	Both indoor & outdoor	Phantom 4 Pro	System design is evaluated for one drone only. LoS channel is considered between UAV & receiver for outdoor scenario.	15 m & 3 km.	Detection rate (Indoor) = 90% Detection rate (Outdoor) = 82%.
[53]	Ground	Simulation	Ideal	Ideal AWGN channel model	Applicable to tele-operated drones only. Noiseless case is considered.	Not specified	Detection performance improved by 6.5 dB & BER level of 10^4 for compression ratio of 4.
[54]	Ground	Experimental	Indoor	DJI Phantom 4 Pro v2.0, DJI Mavic Air, DJI Matrice 600 Pro	The algorithm's resilience in extreme conditions (e.g., dense urban areas, harsh weather) is unexplored.	Not specified	Accuracy rate of 84% and 80% for 2 UAVs (Multi-drone case).
[55]	Ground	Experimental	Not specified	15 different UAVs	Identical RF signals from UAVs cause their classification performance to deteriorate.	Not specified	UAV classification accuracy = 97.25%

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Table 8 (continued).

Ref.	Surveillance station	Experimental/ Simulation	Environment	Drones/ Controllers	Constraints	Range	Outcome
[56]	Ground	Experimental	Both indoor & outdoor	DJI M100, DJI M600, DJI Phantom 3 and DJI Mavic Pro 2.	Approach is dependent upon specific drone models, their signal characteristics (preamble portion of the packet)	–	UAV model classification accuracy = 98.2%
[57]	Ground	Experimental	Indoor	9 different drones	Does not cover outdoor scenarios.	7 m	Detection probability = 96% at –3 dB SNR
[58]	Ground	Experimental	Hilly terrain suburban	Futaba T8J drone radio controller	Only single drone radio controller is considered.	Upto 512 m	Detection probability = 90% at –4 dB SNR
[59]	Ground moving	Numerical analysis	Outdoor	UAV as random point uniformly distributed.	Only single drone radio controller is considered.	Upto 5000 m	Detection probability = 70% at 1000 m
[60]	Ground	Experimental	Both indoor & outdoor	DJI Phantom 4Pro, DJI Inspire Two, Syma-X8hw, Syma-X25pro, and Hm UAVs.	Prone to false UAV detection.	300 m	Recognition rate upto 100%
[61]	Ground	Numerical analysis	Suburban and urban	Generic drone	Fixed drone location is considered.	1 km × 1 km	Detection probability = 94% at 600 m drone height.
[62]	Ground	Experimental	Outdoor	Phantom 4, Inspire, Matrice 600, Mavic Pro 1, FPV RC drone mini quadcopter, and Iris FS-TH9x.	The system is tested for shorter distances only.	8–12 m	UAV/UAV controller detection accuracy = 89.52%.
[63]	Ground	Experimental	Outdoor rural	7 different UAVs	Short distance testing only.	Radius of 40 m	v-measure score = 0.620 with detection in 0.280 seconds
[64]	Ground	Experimental	Both indoor & outdoor	(DJI Phantom 4 Pro v2.0)	Only single UAV is considered.	2 km	UAV recognition > 95% for all classifiers in outdoor.
[65]	Ground	Experimental	Outdoor	5 different drones and their controllers.	Interferences are not considered.	50 m	UAV recognition accuracy > 95% when SNR ≥ 5 dB
[66]	Ground	Experimental	Indoor	Parrot Bebop, Parrot AR Drone, DJI Phantom 3	Dataset specific model evaluation.	Not specified	Detection accuracy = 99.85%, classification accuracy = 98.53%, operating mode recognition accuracy = 95.33%.
[67]	Ground	Experimental	Indoor	15 different UAV controllers.	Short distance testing only.	1–5 m	Classification accuracy > 98% at SNR = –15 dB
[68]	Ground	Experimental	Both indoor & outdoor	DJI Spark and Tello	Limited analysis from interference and UAV variability.	Not specified	Identification accuracy: FD = 100%, AIB = 95.00%, and SIB = 93.50%.
[69]	Ground	Experimental	Outdoor	DJI spark UAV	Only single UAV is considered.	Not specified	Average detection accuracy of combined features = 95.58%
[70]	Ground	Experimental	Indoor	Parrot Bebop, Parrot AR Drone, and DJI Phantom 3.	Dataset does not cover varying drone speed and distance from the RF receiver.	Not specified	Detection accuracy = 99.8%, Type identification accuracy = 94.5%

Table 9

Summary of deployment and performance analysis of WiFi-based drone detection system.

Ref.	Surveil- lance station	Experimen- tal/ Simulation	Environment	Drones/ Controllers	Constraints	Range	Outcome
[15]	Ground	Experimental	Residential	DJI Phantom 3 & Parrot Bebop	Detection is restricted to an attack scenario comprised of launching, surveillance and escaping. Drone is in proximity. Not effective in NLoS scenario.	Upto 62 m in LoS	Presenting the pictorial analysis of Standard deviation changes.
[22]	Ground	Experimental	Working	Parrot AR.Drone 2.0 UAV & its older model, and a Parrot Jumping Race Max terrestrial drone	Can only detect the WiFi operated drones. System performance need to be evaluated for multiple drones. Range restrictions up to WiFi coverage.	WiFi coverage	Optimum classifier: Random tree (Precision = 96%)
[71]	Ground	Experimental	Working	Parrot AR.Drone 2.0	Can only detect the WiFi operated drones. Analysis of encrypted connection between drone and its controller is not highlighted.	WiFi coverage	Optimum classifier: Random tree (TPR = 97%)
[72]	Ground	Simulation	Working	Parrot AR.Drone 2.0	Need prior knowledge of true drone patterns and background patterns. Only one drone was employed to create fake signatures & system evaluation.	WiFi coverage	Detection output < 50%
[73]	Ground	Experimental	Campus	Bebop 1, Bebop 2, DJI Spark, & DBPower UDI	Communication link between UAV & its controller is mandatory.	WiFi coverage	Accuracy = 85.7–95.2% (0.15–0.35 s)
[74]	Ground	Experimental	Outdoor	Bebop 1, Bebop 2, Spark, UDI, Discovery, Tello, TDR, & Wingsland drones.	Communication link between UAV & its controller is mandatory. It is assumed that system is able to identify the application type with un-encrypted WiFi packets.	Upto 70 m in LoS. Upto 40 m in NLoS.	Accuracy = 85.7–95.2% (0.15–0.35 s). Operation mode identification accuracy is 88.5–98.2%.
[75]	Ground	Experimental	Indoor	3DR SOLO drone	Valid only for tele-operated drones. Trade-off between detection performance and delay	WiFi coverage	Identification rate = 93% (3.71 s)
[76]	Ground	Experimental	Six different indoor & outdoor scenarios	3DR SOLO drone & all with same OS.	Communication link between UAV & its controller is mandatory. Drone movement identification for only one drone is considered.	Upto 200 m (outdoor)	Detection performance = 99.8% (0.28 s) Drone movement identification performance = 95% (1.5 s)
[77]	Ground	Experimental	Campus	UPair One	Participation involvement is mandatory. A database is mandatory for MAC address and SSIDs data of all type of drones. Not applicable to customized drones.	Upto 350 m	Detection output = 86.7%.
[78]	Ground	Experimental	Outdoor	DJI Spark, Parrot Bebop, DJI Mavic Pro, and DJI Mavic Air	Detection during video transmission only.	Wi-Fi coverage	Detection output = 86.7%.

boards. As expected, it is also observed that the achieved system performance in terms of detection rate and range proves higher when employing sophisticated hardware with respect to cost-effective solutions, inevitably increasing deployment costs. Therefore, a trade-off must be ensured in the design of suitable and effective drone surveillance systems between costs and performance.

Table 10

Summary of deployment and performance analysis of RF-/WiFi-based drone tracking system.

Ref.	Surveillance station	Source signal	Experimental/ Simulation	Environment	No. of Receivers	Constraints	Range	Outcome
[34]	Ground	RF	Experimental	Outdoor campus	Two	Must be integrated with drone detection system. Only LoS analysis. Limited to angle-side-angle problem. AoA measurement based tracking analysis only.	30–150 m Altitude: 20 m	Pictorial representation of tracking result
[35]	Aerial	RF	Simulations	Not Specified	Three	RSS data is continuously available from the target. Path loss exponent is identical for all links. Obstacle avoidance for UAV swarm is not dealt.	141 m	Tracking error < 10 m
[36]	Aerial	RF	Simulations	Not Specified	Two	Max speed of UAV swarm is 5 m/s. Higher computational complexity. Obstacle free path for UAV swarm.	–	MSE < 50.
[37]	Ground	RF	Experimental	Outdoor	Four	Limited to LoS experiments. Noise co-variance elements are fixed.	1–2 km	Error < 60 m.
[38]	Aerial	RF	Simulations	Ideal	Three - Four	Limited to areas with cellular coverage. Minimum of four base stations are required.	–	Velocity error order of 1 ms^{-1} .
[39]	Ground	RF	Simulations	Indoor	Four	RSSI data can be affected by environmental conditions. Short range experiments. Lacks extensive real-world testing.	30 m	–

One of the main limiting factors emerged from the reviewed studies lies in the utilization of remotely operated drones. Almost all of the systems are designed to handle these types of drones only. Moreover, concerning the WiFi approach, the constraint becomes more stringent, since only WiFi-operated drones can be considered for detection. Also, most of the proposed systems are able to detect the drone only when communicating with its controller without any security protocol. On the contrary, few systems are designed to identify drones with encrypted or secure communication both in the WiFi and RF category. Nonetheless, as discussed previously, anti-drone systems designed so far cannot be considered optimal yet, since their performance is still influenced by multiple parameters and implementation choices. Based on the considered type of surveillance system, several trade-offs have to be made among different parameters in order to ensure satisfactory performance results.

6.2. Drone tracking part

As discussed above, the performance of state-of-the-art tracking systems is still not generalized, hence depending upon different parameters. For instance, in [34] measurements were collected for two types of drones only, and experiments were carried out in open LoS scenario. Further, only simulation-based modeling and performance tests were analyzed in [35,36]. It is also noted that the proposed systems have used sampling-based techniques and algorithms that are usually dependent upon the environment and the tracking objective. Literature in which the drone tracking task is performed with ML/DL algorithms is missing, as reported in Table 7. Moreover, reviewed studies are mainly based upon the use of surveillance UAVs. In this connection, implementing UAV swarms for tracking tasks requires efficient coordination between UAVs and ground stations, and different architectures and tracking schemes can be considered for such task.

Concerning the nature of drone surveillance system, *i.e.*, ground or aerial, it could be mentioned that the type of monitoring station usually depends upon the considered surveillance zone: a small area (e.g., consisting of few buildings) vs. the whole urban area. If the focus is in a short distance range, a ground station might be sufficient, otherwise aerial monitoring devices are required to cover the entire surveillance area. In any case, deployment issues are different when considering the two cases. As far as aerial stations are concerned, general architectural issues to be taken into account are related to the number of devices to be deployed, the type and coordination of communication among nodes, their availability and operational status. Keeping in view the discussion regarding performance evaluation and limitations, it is important to notice that the anti-drone systems proposed so far are particularly effective in the specific conditions for which they have been designed, nonetheless their performance could be

affected when tested in different conditions and scenarios. The concept of generalization cannot be applied yet, as multiple factors such as the nature of receivers, moving/stationary monitoring stations, and the potential intermittent nature of targets significantly impact system performance.

7. Future research directions

Literature works on drone surveillance systems focusing on drone detection and tracking based upon RF/WiFi techniques have been reviewed and discussed so far. Based upon the aforementioned analysis, in the following possible future directions and trends have been identified and analyzed separately for the detection and tracking tasks, keeping in view the context of the studies covered in this SLR.

7.1. Drone detection part

In this section, some interesting research trends and directions concerning drone detection are identified, which we consider worth exploring.

- **Threats identification:** In some scenarios and situations, if the final aim is not to neutralize the detected intruder drone by applying interdiction techniques, yet it is required to intercept the drone's communication with its controller. In addition to eavesdropping the drone communication, it might be important to decode and interpret the captured traffic, so that prior information on possible attacks may be obtained in advance. In this context, few studies have been presented aimed at decoding the collected signals, both for RF and WiFi approaches, as can be noted from [Tables 5](#) and [6](#). Indeed, such task is quite challenging due to the variety of drones available on the market and the different communication protocols used by UAVs and their controllers.
- **Range of detection:** A large majority of the research work so far concerns drone detection, considering a surveillance range up to 3 km in RF-based approach; however this can be achieved only when employing high gain receivers, which causes the cost of the system to increase significantly. Therefore, a cost-effective anti-drone system providing optimum performance in long-range scenarios still needs to be designed. Furthermore, the range of detection in the category of WiFi is normally limited to WiFi range frequencies. Instead of this, detecting drones obtained by implementing WiFi technique is inspiring and promising towards its real deployment at the end-user side. The effectiveness of such systems could be improved by devising ad-hoc networks of WiFi monitoring devices, deployed on the field in order to increase the coverage area and the correct detection rate of intruders.
- **Detection of autonomous drones:** it is important to consider that the intruder drone might not be remotely operated by a user, yet it could be devised for autonomous flight. This scenario poses a significant challenge for RF and WiFi techniques-based detection systems, since the large majority of the proposed systems perform the detection task based on drones' communication signals. In this perspective, the detection of autonomous UAVs has been implemented so far for known drones only, based on prior information about the target. The detection task becomes more challenging when an unknown drone enters the surveillance area, and this specific scenario is still to be investigated.
- **Building of large database:** To design a hybrid system that may not only be able to detect different types of drones but also to interpret the communication between the drone and its controller is a quite interesting and current research topic. A large database is usually required normally to identify and classify multiple types of drones. Therefore, if a system must be able also to decode and interpret the drone's transmissions, an even larger amount of data is needed to be stored for anti-drone system effectiveness. In this connection, [\[47,65\]](#) created database including three and five UAVs, respectively. In this perspective, the robust and timely detection and accurate classification of dozens of drones and their statuses pose significant challenges that need to be addressed.
- **NLoS conditions:** Detection of drones individually for LoS and NLoS conditions was discussed in [\[15,46\]](#). Relevant issues concerning these scenarios were also analyzed in [\[46\]](#). However, performance studies need to be carried out for different outdoor environments, especially with NLoS conditions and in the presence of interference, both for RF and WiFi techniques. This could improve detection system in terms of compactness, robustness and versatility.
- **Recognition of different types of drones:** 3DR Solo drone with ArduCopter 3.3 firmware was used in the experimental setup in [\[76\]](#), and authors claim that the system can detect all drones with the same operating system. Although this work considers the detection of multiple drones of the same type, recognizing different types of drones is a matter still uncovered. Since drones manufactured by different companies are implemented with specific and sometimes proprietary operating systems and communication protocols, a suitable anti-drone system needs to be designed to be able to identify not only a single specific type of drone, but a set of devices produced by different manufacturers. Therefore, optimum ML/DL algorithms both in the category of RF and WiFi techniques should be investigated, to generalize the effectiveness and applicability of drone surveillance systems.
- **Detection of multiple drones:** Almost all the research papers discussed and reviewed in this SLR tackle the problem of detecting only one drone in experimental scenarios. However, not necessarily the attack planning is performed by one single drone. A larger number of UAVs could also be employed for such task. Hence, the detection of multiple drones simultaneously, both in the category of RF and WiFi approaches, is a primary challenge that needs to be addressed.

- **Specific Emitter Identification (SEI):** Measurements of external features of RF signals are referred to as RF fingerprints, and for any given signal, SEI designates an unique transmitter. For example, DSSS-QPSK signals and OFDM-BPSK from WLAN cards and digital radios, respectively, can be used to evaluate the performance of SEI [87]. Advanced techniques need to be inspected, analyzed, and evaluated concerning SEI in the context of drone surveillance systems.
- **Detection with intermittent/fake data/covert transmission:** In the surveyed literature, drone detection with encrypted WiFi traffic has been dealt with, detection in a covert transmission scenario is also analyzed numerically in the RF category, and intelligent adversary techniques are also proposed, by generating fake data to deceive the surveillance systems in the WiFi-based approach. In this connection, an intelligent adversary may also operate with intermittent transmissions with the same deceiving intentions. Therefore, intelligent and effective system modeling must be explored and designed to deal with such an intelligent adversary both in the RF and WiFi scenarios.
- **Deployment of aerial stations:** When deployed for long-range detections of intruders, drones face a number of challenges. First, their high mobility, whether due to wind, altitude changes, or rapid movement, demands reliable communication links. Second, dynamic relaying is critical: drones frequently serve as network relays, requiring fast handoffs and relay management. Third, balancing throughput (data rate) and latency (reaction time) is critical for effective communication over long distances. Fourth, energy efficiency is critical, as drones rely on limited battery capacity for long missions. Fifth, maintaining good signal coverage and reliable communication in the face of interference and noise is difficult. Finally, security and privacy considerations are critical, particularly when communicating sensitive information.
- **Remote ID adaptability:** The remote ID standard identifies legal and malicious drones in an area, but attacks against legal drones hinder privacy safeguards. Intentional circumvention by attackers poses significant challenges, including privacy-aware sharing of sensitive information, making it difficult to detect unauthorized drones. Sophisticated methods and a comprehensive regulatory framework are needed to address these issues.

7.2. Drone tracking part

Different sub-tasks and challenges can be identified for the implementation of the anti-drone systems proposed in the literature. In this connection, it is worth noting that, while RF-based tracking of intruders has been performed in the literature, a proper solution for the same task in WiFi-based systems still needs to be investigated. Moreover, a few additional issues regarding drone tracking systems are highlighted below.

- **Trajectory prediction:** Trajectory prediction based upon the study of the history of drone movements could be an interesting feature for a drone surveillance system, either consisting of aerial or ground monitoring station. Although this is a rather challenging task due to the non-cooperative nature of the malicious drone, including such capability using suitable techniques could be a significant added value for a UAV surveillance system.
- **Intermittent, covert & encrypted transmission:** The intruder UAV not necessarily transmits signals continuously, and in addition such communication might be encrypted. A drone operator always focuses on escaping and securing the drone from any surveillance station deployed, and this can be achieved by producing intermittent signals and using covert communication between drone and controller. Therefore, an important feature for an effective surveillance system consists in taking into account possible intentional alteration of the transmitted signals, both in terms of intermittent communications and ciphered messages. In particular, anti-drone systems should be able to detect and track UAVs also in case of such intentional countermeasures employed by the attacker.
- **Implementation of ML algorithms:** It is observed from the literature that the majority of the work proposed so far is based upon non-ML algorithms. Their applicability in the framework of amateur drone tracking is lacking in the literature. Therefore, it would be interesting to explore the feasibility of such approaches to be implemented for drone tracking systems.
- **Tracking of multiple drones:** Similarly to the detection and localization of multiple drones, tracking of more than one drone is also necessary. Since this is an unexplored topic yet, solutions need to be investigated to efficiently perform this task.
- **Tracking via ground base stations:** Mainly, the work presented in the literature so far is based upon aerial nodes, rather than ground stations. However, this choice could depend upon the location where the surveillance system needs to be deployed. For instance, a small monitoring area such as a building, may not require a UAV or UAV swarm to be covered. Hence, tracking systems consisting of ground surveillance stations still need to be further explored. In addition, the anti-drone system is still devoid of a combined ground and aerial surveillance station, which would significantly improve its tracking and detection capabilities.
- **Practical deployment:** The practical or real-world deployment of drone tracking systems introduces additional challenges such as dynamic environments, varying terrain, and interference from other wireless devices, which must be considered alongside controlled and simulation based experiments for accurate and reliable evaluations.

In addition to the challenges discussed above, depending upon the type of monitoring station (aerial or ground), other types of issues can be experienced while designing a drone tracking system. For instance, in the case of aerial surveillance, possible challenges could consist in: (i) deployment of standalone UAVs or in coordination with a ground base station, (ii) deployment of single or multiple UAVs, (iii) communication and coordination within the UAV swarm, (iv) choice of a suitable architecture for multiple UAVs, (v) efficient data collection from targeted drone in dynamic environments, (vi) smart path planning for UAV(s), (vii) obstacle & collision avoidance strategies, and (viii) accurate tracking within minimum time.

8. Conclusion

A systematic review work concerning RF/WiFi-based drone surveillance systems has been presented in the perspective of detection and tracking systems. The implemented systems for detecting single and multiple drones, based on simulations and experimental work, have been reviewed in detail, along with the discussion related to the associated challenges. For what concerns detection, both RF and WiFi-based systems have been proposed and implemented in the literature. On the contrary, as far as tracking is concerned, the published work is primarily based on RF techniques only, and both ground- and aerial-based surveillance systems have been analyzed. Furthermore, drone tracking solutions based on a WiFi approach have not been implemented yet. A thorough discussion on the results provided by the surveyed works has been presented, for both the detection and tracking tasks. Finally, in light of the reviewed work, open challenges and future research trends for drone detection and tracking have been outlined.

CRedit authorship contribution statement

Igor Bisio: Conceptualization, Investigation, Supervision, Validation, Writing – original draft, Writing – review & editing. **Chiara Garibotto:** Conceptualization, Investigation, Supervision, Validation, Writing – original draft, Writing – review & editing. **Halar Haleem:** Conceptualization, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Fabio Lavagetto:** Conceptualization, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing. **Andrea Sciarrone:** Conceptualization, Investigation, Methodology, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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